

Physics 523 HW 2

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I use the conventions and notation in my QED & Real Scalar Field Notes, which are mostly based on Schwartz. I usually use p for the magnitude of the momentum 3-vector, $p = |\mathbf{p}|$, so $p^2 = \mathbf{p} \cdot \mathbf{p}$. But for this homework I will use p^2 for the dot product of the momentum 4-vector: $p^2 = \vec{p} \cdot \vec{p} = \vec{p}^2 = (p^0)^2 - \mathbf{p}^2$.

1. PHOTON SELF ENERGY

I follow the calculation in Schwartz Sec. 16.2.2. I draw the photon self energy diagram, $\star$, in Fig. 1. I took care of momentum conservation in the diagram so I don't need vertex delta functions. The external photons are off-shell. Since I am following Schwartz I adopt his convention that in loops, momentum coincides with particle flow.

$$\star = -\frac{(-ie)^2 S}{(2\pi)^4} \int d^4 q \text{Tr} \left[\gamma^\mu i \frac{\not{q} - \not{p} + m}{(\vec{q} - \vec{p})^2 - m^2 + i\epsilon} \gamma^\nu i \frac{\not{q} + m}{q^2 - m^2 + i\epsilon} \right] \quad (1)$$

I show the symmetry factor, S , combinatorics in Fig. 2.

$$S = \frac{2}{2!} = \frac{2}{2} = 1 \quad (2)$$

$$\star = -\frac{e^2}{(2\pi)^4} \int d^4 q \frac{\text{Tr}[\gamma^\mu (\not{q} - \not{p} + m) \gamma^\nu (\not{q} + m)]}{[(\vec{q} - \vec{p})^2 - m^2 + i\epsilon](q^2 - m^2 + i\epsilon)} \quad (3)$$

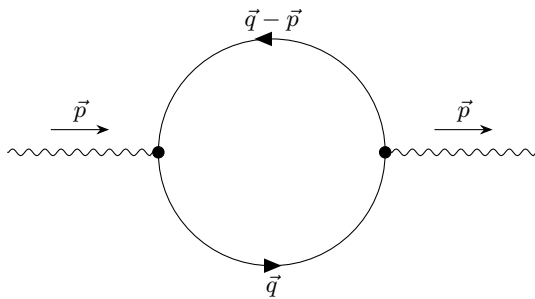


FIG. 1. $\star$

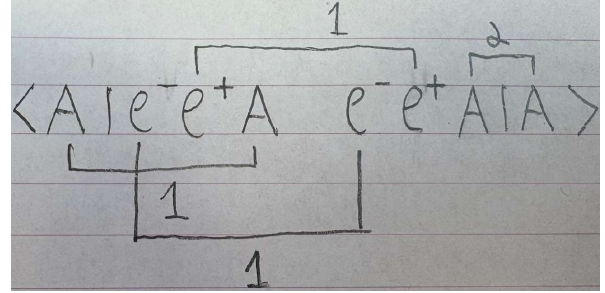


FIG. 2. Photon self energy symmetry factor combinatorics.

$$\star = -\frac{e^2}{(2\pi)^4} \int d^4 q \frac{N}{[(\vec{q} - \vec{p})^2 - m^2 + i\epsilon](q^2 - m^2 + i\epsilon)} \quad (4)$$

$$N = \text{Tr}[\gamma^\mu (\not{q} - \not{p} + m) \gamma^\nu (\not{q} + m)] \quad (5)$$

$$N = \text{Tr}(\gamma^\mu \not{q} \gamma^\nu \not{q}) - \text{Tr}(\gamma^\mu \not{p} \gamma^\nu \not{q}) + m \text{Tr}(\gamma^\mu \gamma^\nu \not{q}) + m \text{Tr}(\gamma^\mu \not{q} \gamma^\nu) - m \text{Tr}(\gamma^\mu \not{p} \gamma^\nu) + m^2 \text{Tr}(\gamma^\mu \gamma^\nu) \quad (6)$$

$$N = q_\alpha q_\beta \text{Tr}(\gamma^\mu \gamma^\alpha \gamma^\nu \gamma^\beta) - p_\alpha q_\beta \text{Tr}(\gamma^\mu \gamma^\alpha \gamma^\nu \gamma^\beta) + m q_\alpha \text{Tr}(\gamma^\mu \gamma^\alpha \gamma^\nu) + m q_\alpha \text{Tr}(\gamma^\mu \gamma^\alpha \gamma^\nu) - m p_\alpha \text{Tr}(\gamma^\mu \gamma^\alpha \gamma^\nu) + m^2 \text{Tr}(\gamma^\mu \gamma^\nu) \quad (7)$$

$$\text{Tr}(\gamma^\alpha \gamma^\beta \gamma^\mu) = 0 \quad (8)$$

$$N = (q_\alpha q_\beta - p_\alpha q_\beta) \text{Tr}(\gamma^\mu \gamma^\alpha \gamma^\nu \gamma^\beta) + m^2 \text{Tr}(\gamma^\mu \gamma^\nu) \quad (9)$$

$$\text{Tr}(\gamma^\mu \gamma^\nu) = 4\eta^{\mu\nu} \quad (10)$$

$$\text{Tr}(\gamma^\mu \gamma^\alpha \gamma^\nu \gamma^\beta) = 4(\eta^{\mu\alpha} \eta^{\nu\beta} - \eta^{\mu\nu} \eta^{\alpha\beta} + \eta^{\mu\beta} \eta^{\alpha\nu}) \quad (11)$$

$$\frac{N}{4} = (q_\alpha q_\beta - p_\alpha q_\beta)(\eta^{\mu\alpha} \eta^{\nu\beta} - \eta^{\mu\nu} \eta^{\alpha\beta} + \eta^{\mu\beta} \eta^{\alpha\nu}) + m^2 \eta^{\mu\nu} \quad (12)$$

$$\frac{N}{4} = q^\mu q^\nu - p^\mu q^\nu + \eta^{\mu\nu} \vec{q} \cdot \vec{p} - \eta^{\mu\nu} q^2 + q^\mu q^\nu - q^\mu p^\nu + m^2 \eta^{\mu\nu} \quad (13)$$

$$\frac{N}{4} = 2q^\mu q^\nu - p^\mu q^\nu - q^\mu p^\nu + (m^2 + \vec{q} \cdot \vec{p} - q^2) \eta^{\mu\nu} \quad (14)$$

I define

$$i\Pi^{\mu\nu} = \text{✱} \implies \Pi^{\mu\nu} = -i\text{✱}. \quad (15)$$

$$\Pi^{\mu\nu} = \frac{ie^2}{(2\pi)^4} \int d^4 q \frac{N}{[(\vec{q} - \vec{p})^2 - m^2 + i\epsilon](q^2 - m^2 + i\epsilon)} \quad (16)$$

$$\Pi^{\mu\nu} = \frac{4ie^2}{(2\pi)^4} \int d^4 q \frac{2q^\mu q^\nu - p^\mu q^\nu - q^\mu p^\nu + (m^2 + \vec{q} \cdot \vec{p} - q^2) \eta^{\mu\nu}}{[(\vec{q} - \vec{p})^2 - m^2 + i\epsilon](q^2 - m^2 + i\epsilon)} \quad (17)$$

By Lorentz invariance

$$\Pi^{\mu\nu} = p^2 F_1(p^2, m^2) \eta^{\mu\nu} + F_2(p^2, m^2) p^\mu p^\nu, \quad (18)$$

where F_1 and F_2 are called form factors. To simplify the integral I introduce Feynman parameters.

$$\frac{1}{AB} = \int_0^1 \frac{dx}{[A + (B - A)x]^2} = \int_0^1 \frac{dx}{D^2} \quad (19)$$

$$A = (\vec{q} - \vec{p})^2 - m^2 + i\epsilon, \quad B = q^2 - m^2 + i\epsilon \quad (20)$$

$$D = A + (B - A)x = (\vec{q} - \vec{p})^2 + (2\vec{q} \cdot \vec{p} - p^2)x - m^2 + i\epsilon \quad (21)$$

$$D = q^2 + 2(x - 1)\vec{p} \cdot \vec{q} + (1 - x)p^2 - m^2 + i\epsilon \quad (22)$$

Completing the square means

$$ax^2 + bx + c = a(x + d)^2 + e, \quad d = \frac{b}{2a}, \quad e = c - \frac{b^2}{4a}. \quad (23)$$

I apply this to D by taking

$$x \rightarrow \vec{q}, \quad a \rightarrow 1, \quad b \rightarrow 2(x - 1)\vec{p}, \quad c \rightarrow (1 - x)p^2 - m^2 + i\epsilon. \quad (24)$$

$$D = [\vec{q} + (x - 1)\vec{p}]^2 + x(1 - x)p^2 - m^2 + i\epsilon \quad (25)$$

$$\Delta = m^2 - x(1 - x)p^2 \quad (26)$$

$$D = [\vec{q} + (x - 1)\vec{p}]^2 - \Delta + i\epsilon \quad (27)$$

$$\Pi^{\mu\nu} = \frac{4ie^2}{(2\pi)^4} \int d^4 q \frac{2q^\mu q^\nu - p^\mu q^\nu - q^\mu p^\nu + (m^2 + \vec{q} \cdot \vec{p} - q^2) \eta^{\mu\nu}}{AB} \quad (28)$$

$$\Pi^{\mu\nu} = \frac{4ie^2}{(2\pi)^4} \int d^4 q \int_0^1 dx \frac{2q^\mu q^\nu - p^\mu q^\nu - q^\mu p^\nu + (m^2 + \vec{q} \cdot \vec{p} - q^2) \eta^{\mu\nu}}{D^2} \quad (29)$$

$$\Pi^{\mu\nu} = \frac{4ie^2}{(2\pi)^4} \int d^4 q \int_0^1 dx \frac{2q^\mu q^\nu - p^\mu q^\nu - q^\mu p^\nu + (m^2 + \vec{q} \cdot \vec{p} - q^2) \eta^{\mu\nu}}{[\vec{q} + (x - 1)\vec{p}]^2 - \Delta + i\epsilon]^2} \quad (30)$$

$$\Pi^{\mu\nu} = \frac{4ie^2}{(2\pi)^4} \int_0^1 dx \int d^4 q \frac{2q^\mu q^\nu - p^\mu q^\nu - q^\mu p^\nu + (m^2 + \vec{q} \cdot \vec{p} - q^2) \eta^{\mu\nu}}{[\vec{q} + (x - 1)\vec{p}]^2 - \Delta + i\epsilon]^2} \quad (31)$$

I shift the integrand.

$$\begin{aligned} \vec{k} &= \vec{q} - (1 - x)\vec{p}, & d^4 k &= d^4 q, \\ \vec{q} &= \vec{k} + (1 - x)\vec{p}, & \implies q^\mu &= k^\mu + (1 - x)p^\mu \\ q^2 &= k^2 + (1 - x)^2 p^2 + 2(1 - x)\vec{p} \cdot \vec{k} \end{aligned} \quad (32)$$

$$\begin{aligned} \Pi^{\mu\nu} &= \frac{4ie^2}{(2\pi)^4} \int_0^1 dx \int \frac{d^4 k}{(k^2 - \Delta + i\epsilon)^2} \\ &\quad [2k^\mu k^\nu - 2x(1 - x)p^\mu p^\nu + \cancel{(1 - 2x)(k^\mu p^\nu + p^\mu k^\nu)} \\ &\quad + [m^2 - k^2 + x(1 - x)p^2 - \cancel{(1 - 2x)\vec{p} \cdot \vec{k}}] \eta^{\mu\nu}] \end{aligned} \quad (33)$$

The pk terms cancel because the integrand is odd.

$$\begin{aligned} \Pi^{\mu\nu} &= \frac{4ie^2}{(2\pi)^4} \int_0^1 dx \int \frac{d^4 k}{(k^2 - \Delta + i\epsilon)^2} \\ &\quad [2k^\mu k^\nu - 2x(1 - x)p^\mu p^\nu + [m^2 - k^2 + x(1 - x)p^2] \eta^{\mu\nu}] \end{aligned} \quad (34)$$

$$\begin{aligned} \Pi^{\mu\nu} &= \frac{4ie^2}{(2\pi)^4} \int_0^1 dx \int \frac{d^4 k}{(k^2 - \Delta + i\epsilon)^2} \\ &\quad [2k^\mu k^\nu - 2x(1 - x)p^\mu p^\nu - (k^2 + \Delta - 2m^2) \eta^{\mu\nu}] \end{aligned} \quad (35)$$

$$\Pi^{\mu\nu} = \frac{4ie^2}{(2\pi)^4} \int_0^1 dx \mathbb{I} \quad (36)$$

$$\mathbb{I} = \int d^4k \frac{2k^\mu k^\nu - 2x(1-x)p^\mu p^\nu - (k^2 + \Delta - 2m^2)\eta^{\mu\nu}}{(k^2 - \Delta + i\epsilon)^2} \quad (37)$$

$$\mathbb{I} = 2\mathbb{I}_1 - \eta^{\mu\nu}\mathbb{I}_2 - [(\Delta - 2m^2)\eta^{\mu\nu} + 2x(1-x)p^\mu p^\nu]\mathbb{I}_3 \quad (38)$$

$$\mathbb{I}_1 = \int d^4k \frac{k^\mu k^\nu}{(k^2 - \Delta + i\epsilon)^2}, \quad \mathbb{I}_2 = \int d^4k \frac{k^2}{(k^2 - \Delta + i\epsilon)^2},$$

$$\mathbb{I}_3 = \int \frac{d^4k}{(k^2 - \Delta + i\epsilon)^2} \quad (39)$$

I use power counting to see how these integrals diverge,

$$\mathbb{I}_1 \sim \mathbb{I}_2 \sim \int_1^\Lambda dk k \sim \Lambda^2 \quad \mathbb{I}_3 = \int_1^\Lambda \frac{dk}{k} = \ln \Lambda, \quad (40)$$

where $\Lambda \rightarrow \infty$. I use dimensional regularization.

$$\Pi^{\mu\nu}(d) = \frac{4ie^2}{(2\pi)^d} \mu^{4-d} \int_0^1 dx \mathbb{I}(d) \quad (41)$$

$$\mathbb{I}(d) = 2\mathbb{I}_1(d) - \eta^{\mu\nu}\mathbb{I}_2(d) - [(\Delta - 2m^2)\eta^{\mu\nu} + 2x(1-x)p^\mu p^\nu]\mathbb{I}_3(d) \quad (42)$$

$$\mathbb{I}_1(d) = \int d^d k \frac{k^\mu k^\nu}{(k^2 - \Delta + i\epsilon)^2},$$

$$\mathbb{I}_2(d) = \int d^d k \frac{k^2}{(k^2 - \Delta + i\epsilon)^2}, \quad \mathbb{I}_3(d) = \int \frac{d^d k}{(k^2 - \Delta + i\epsilon)^2} \quad (43)$$

I use Schwartz Eq. B.51

$$\int d^d k \frac{k^\mu k^\nu}{(k^2 + a)^n} = \frac{\eta^{\mu\nu}}{d} \int d^d k \frac{k^2}{(k^2 + a)^n}. \quad (44)$$

$$\mathbb{I}_1(d) = \int d^d k \frac{k^\mu k^\nu}{(k^2 - \Delta + i\epsilon)^2} =$$

$$\frac{\eta^{\mu\nu}}{d} \int d^d k \frac{k^2}{(k^2 - \Delta + i\epsilon)^2} = \frac{\eta^{\mu\nu}}{d} \mathbb{I}_2(d) \quad (45)$$

$$\mathbb{I}(d) = \eta^{\mu\nu} \left(\frac{2}{d} - 1 \right) \mathbb{I}_2(d) - [(\Delta - 2m^2)\eta^{\mu\nu} + 2x(1-x)p^\mu p^\nu]\mathbb{I}_3(d) \quad (46)$$

I use two integrals I derived in my Real Scalar Field Notes.

$$\int \frac{d^d p}{(p^2 - \Delta + i\epsilon)^2} = \frac{i\pi^{d/2}}{\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right)$$

$$\int d^d p \frac{p^2}{(p^2 - \Delta + i\epsilon)^2} = \frac{d\pi^{d/2}}{2i\Delta^{1-d/2}} \Gamma\left(\frac{2-d}{2}\right) \quad (47)$$

$$\mathbb{I}_2(d) = \frac{d\pi^{d/2}}{2i\Delta^{1-d/2}} \Gamma\left(\frac{2-d}{2}\right), \quad \mathbb{I}_3(d) = \frac{i\pi^{d/2}}{\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right) \quad (48)$$

$$\mathbb{I}(d) = \eta^{\mu\nu} \left(\frac{2}{d} - 1 \right) \frac{d\pi^{d/2}}{2i\Delta^{1-d/2}} \Gamma\left(\frac{2-d}{2}\right) -$$

$$[(\Delta - 2m^2)\eta^{\mu\nu} + 2x(1-x)p^\mu p^\nu] \frac{i\pi^{d/2}}{\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right) \quad (49)$$

$$\mathbb{I}(d) = \frac{\pi^{d/2}}{i\Delta^{1-d/2}} \left\{ \eta^{\mu\nu} \left[\left(\frac{2}{d} - 1 \right) \frac{d}{2} \Gamma\left(\frac{2-d}{2}\right) + \frac{(\Delta - 2m^2)}{\Delta} \Gamma\left(\frac{4-d}{2}\right) \right] \right.$$

$$\left. + \frac{2}{\Delta} x(1-x)p^\mu p^\nu \Gamma\left(\frac{4-d}{2}\right) \right\} \quad (50)$$

$$\mathbb{I}(d) = \frac{\pi^{d/2}}{i\Delta^{1-d/2}} \left\{ \frac{2}{\Delta} x(1-x)p^\mu p^\nu \Gamma\left(2 - \frac{d}{2}\right) + \right.$$

$$\left. \eta^{\mu\nu} \left[\left(1 - \frac{d}{2}\right) \Gamma\left(1 - \frac{d}{2}\right) + \left(1 - 2\frac{m^2}{\Delta}\right) \Gamma\left(2 - \frac{d}{2}\right) \right] \right\} \quad (51)$$

$$\Gamma(1+x) = x\Gamma(x) \quad (52)$$

$$\mathbb{I}(d) = \frac{2\pi^{d/2}}{i\Delta^{1-d/2}} \left[\frac{x}{\Delta} (1-x)p^\mu p^\nu + \eta^{\mu\nu} \left(1 - \frac{m^2}{\Delta}\right) \right]$$

$$\Gamma\left(2 - \frac{d}{2}\right) \quad (53)$$

$$\mathbb{I}(d) = \frac{2\pi^{d/2}}{i\Delta^{2-d/2}} [x(1-x)p^\mu p^\nu + \eta^{\mu\nu}(\Delta - m^2)] \Gamma\left(2 - \frac{d}{2}\right) \quad (54)$$

$$\mathbb{I}(d) = 2i\pi^{d/2} (\eta^{\mu\nu} p^2 - p^\mu p^\nu) \Gamma\left(2 - \frac{d}{2}\right) x \frac{1-x}{\Delta^{2-d/2}} \quad (55)$$

$$\Pi^{\mu\nu}(d) = \frac{e^2 \mu^{4-d}}{2^{d-3} \pi^{d/2}} (p^\mu p^\nu - \eta^{\mu\nu} p^2) \Gamma\left(2 - \frac{d}{2}\right)$$

$$\int_0^1 dx x \frac{1-x}{\Delta^{2-d/2}} \quad (56)$$

$$\Pi^{\mu\nu}(d) = \frac{e^2 \mu^{4-d}}{2^{d-3} \pi^{d/2}} (p^\mu p^\nu - p^2 \eta^{\mu\nu}) \Gamma\left(2 - \frac{d}{2}\right)$$

$$\int_0^1 dx \frac{x(1-x)}{[m^2 - x(1-x)p^2]^{2-d/2}} \quad (57)$$

I return to 4D.

$$\Pi^{\mu\nu} = \Pi^{\mu\nu}(4) = \lim_{\epsilon \rightarrow 0} \Pi^{\mu\nu}(4 - \epsilon) \quad (58)$$

Let the limit $\epsilon \rightarrow 0$ be implicit.

$$\Pi^{\mu\nu} = \Pi^{\mu\nu}(4 - \epsilon) \quad (59)$$

$$\Pi^{\mu\nu} = \frac{e^2(2\mu\sqrt{\pi})^\epsilon}{2\pi^2} (p^\mu p^\nu - p^2 \eta^{\mu\nu}) \Gamma\left(\frac{\epsilon}{2}\right) \int_0^1 dx \frac{x(1-x)}{[m^2 - x(1-x)p^2]^{\epsilon/2}} \quad (60)$$

I expand around $\epsilon = 0$.

$$\Pi^{\mu\nu} = \frac{e^2}{2\pi^2} (p^\mu p^\nu - p^2 \eta^{\mu\nu}) \int_0^1 dx x(1-x) \left(2\mu \sqrt{\frac{\pi}{m^2 - x(1-x)p^2}} \right)^\epsilon \Gamma\left(\frac{\epsilon}{2}\right) \quad (61)$$

$$\Pi^{\mu\nu} = \frac{e^2}{2\pi^2} (p^\mu p^\nu - p^2 \eta^{\mu\nu}) \int_0^1 dx x(1-x) \left[\frac{2}{\epsilon} + \ln\left(\frac{\tilde{\mu}^2}{m^2 - x(1-x)p^2}\right) \right] \quad (62)$$

Where

$$\tilde{\mu}^2 = 4\pi e^{-\gamma} \mu^2, \quad (63)$$

where e is Euler's number, not the electric charge, and $\gamma \approx 0.577216$ is the Euler-Mascheroni constant. I define

$$\Pi_2(p^2) = \int_0^1 \frac{dx}{2\pi^2} x(1-x) \left[\frac{2}{\epsilon} + \ln\left(\frac{\tilde{\mu}^2}{m^2 - x(1-x)p^2}\right) \right] \quad (64)$$

The regularized diagram is

$$\Pi^{\mu\nu} = e^2 (p^\mu p^\nu - p^2 \eta^{\mu\nu}) \Pi_2(p^2). \quad (65)$$

2. VERTEX CORRECTION

I follow Schwartz Secs. 17.2 & 19.3. I show the relevant Feynman diagram, $\star$, in Fig. 3. I show the combinatorics for the symmetry factor S in Fig. 4.

$$S = \frac{4 \times 3 \times 2^2}{3!(2!)^3} = 1 \quad (66)$$

Momentum conservation at each vertex implies

$$\vec{p} = \vec{p}_2 - \vec{p}_1. \quad (67)$$

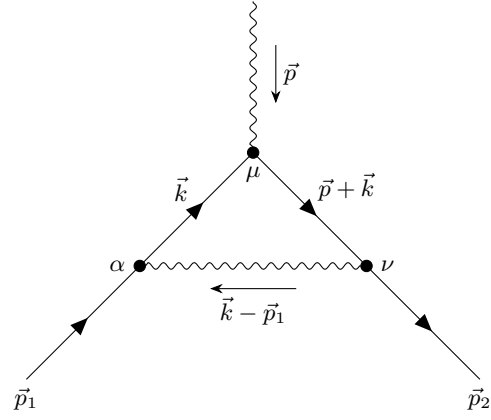


FIG. 3. Feynman diagram $\star$. The external photon is off-shell. I suppress the spin indices.

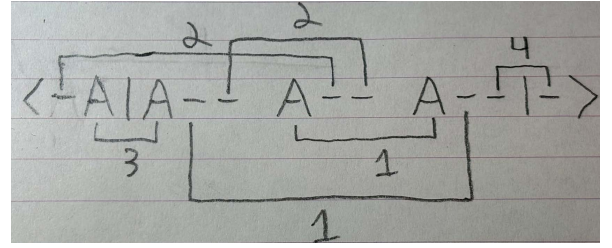


FIG. 4. Vertex correction combinatorics. The key is to recognize that every fermion in the diagram is an electron (-).

The QED Feynman rules yield

$$\star = S \int \frac{d^4 k}{(2\pi)^4} \frac{-i\eta_{\nu\alpha}}{(\vec{k} - \vec{p}_1)^2 + i\epsilon} \bar{u}(\vec{p}_2)(-ie\gamma^\nu) i \frac{\not{p} + \not{k} + m}{(\vec{p} + \vec{k})^2 - m^2 + i\epsilon} (-ie\gamma^\mu) i \frac{\not{k} + m}{k^2 - m^2 + i\epsilon} (-ie\gamma^\alpha) u(\vec{p}_1). \quad (68)$$

I introduce a fictitious photon mass λ to prevent an IR divergence.

$$\star = S \int \frac{d^4 k}{(2\pi)^4} \frac{-i\eta_{\nu\alpha}}{(\vec{k} - \vec{p}_1)^2 - \lambda^2 + i\epsilon} \bar{u}(\vec{p}_2)(-ie\gamma^\nu) i \frac{\not{p} + \not{k} + m}{(\vec{p} + \vec{k})^2 - m^2 + i\epsilon} (-ie\gamma^\mu) i \frac{\not{k} + m}{k^2 - m^2 + i\epsilon} (-ie\gamma^\alpha) u(\vec{p}_1). \quad (69)$$

I use the shorthand

$$u_n = u(\vec{p}_n). \quad (70)$$

$$\star = -\frac{e^3 \eta_{\nu\alpha}}{(2\pi)^4} \int \frac{d^4 k}{(\vec{k} - \vec{p}_1)^2 - \lambda^2 + i\epsilon} \frac{\bar{u}_2 \gamma^\nu (\not{p} + \not{k} + m) \gamma^\mu (\not{k} + m) \gamma^\alpha u_1}{[(\vec{p} + \vec{k})^2 - m^2 + i\epsilon](k^2 - m^2 + i\epsilon)} \quad (71)$$

$$\star = -\frac{e^3}{(2\pi)^4} \int \frac{d^4 k}{(\vec{k} - \vec{p}_1)^2 - \lambda^2 + i\epsilon} \frac{\bar{u}_2 \gamma^\nu (\not{p} + \not{k} + m) \gamma^\mu (\not{k} + m) \gamma_\nu u_1}{[(\vec{p} + \vec{k})^2 - m^2 + i\epsilon](k^2 - m^2 + i\epsilon)} \quad (72)$$

I simplify $\star$ using Feynman parameters.

$$A = k^2 - m^2 + i\epsilon, \quad B = (\vec{p} + \vec{k})^2 - m^2 + i\epsilon, \quad C = (\vec{k} - \vec{p}_1)^2 - \lambda^2 + i\epsilon \quad (73)$$

$$\star = -e^3 \int d^4 k \frac{\bar{u}_2 \gamma^\nu (\not{p} + \not{k} + m) \gamma^\mu (\not{k} + m) \gamma_\nu u_1}{(2\pi)^4 ABC} \quad (74)$$

$$\frac{1}{ABC} = 2 \int_0^1 dx dy dz \frac{\delta(x+y+z-1)}{(xA+yB+zC)^3} \quad (75)$$

$$xA+yB+zC = (x+y+z)(k^2+i\epsilon) - (x+y)m^2 + yp^2 + zp_1^2 + 2(y\vec{p} - z\vec{p}_1) \cdot \vec{k} - z\lambda^2 \quad (76)$$

Because of the delta function in Eq. 75 I can use the identity $x+y+z=1$.

$$D = xA+yB+zC = k^2 + 2(y\vec{p} - z\vec{p}_1) \cdot \vec{k} + yp^2 + zp_1^2 - (x+y)m^2 - z\lambda^2 + i\epsilon \quad (77)$$

I complete the square.

$$ax^2+bx+c = a(x+d)^2+e, \quad d = \frac{b}{2a}, \quad e = c - \frac{b^2}{4a}. \quad (78)$$

$$a \rightarrow 1, \quad b \rightarrow 2(y\vec{p} - z\vec{p}_1), \quad x \rightarrow \vec{k} \\ c \rightarrow yp^2 + zp_1^2 - (x+y)m^2 - z\lambda^2 + i\epsilon \quad (79)$$

$$D = (\vec{k} + y\vec{p} - z\vec{p}_1)^2 + yp^2 + zp_1^2 - (x+y)m^2 - z\lambda^2 + i\epsilon - (y\vec{p} - z\vec{p}_1)^2 \quad (80)$$

$$D = (\vec{k} + y\vec{p} - z\vec{p}_1)^2 + y(1-y)p^2 + z(1-z)p_1^2 + 2yz\vec{p} \cdot \vec{p}_1 - (x+y)m^2 - z\lambda^2 + i\epsilon \quad (81)$$

The external electrons in $\star$ are on-shell so

$$p_1^2 = p_2^2 = m^2. \quad (82)$$

$$D = (\vec{k} + y\vec{p} - z\vec{p}_1)^2 + y(1-y)p^2 + 2yz\vec{p} \cdot \vec{p}_1 + (z-x-y-z^2)m^2 - z\lambda^2 + i\epsilon \quad (83)$$

$$D = (\vec{k} + y\vec{p} - z\vec{p}_1)^2 + y[(1-y)p^2 + 2z\vec{p} \cdot \vec{p}_1] - (1-z)^2 m^2 - z\lambda^2 + i\epsilon \quad (84)$$

I combine Eqs. 67 & 82.

$$p^2 = (\vec{p}_2 - \vec{p}_1)^2 = p_2^2 + p_1^2 - 2\vec{p}_1 \cdot \vec{p}_2 = 2m^2 - 2\vec{p}_1 \cdot \vec{p}_2 \quad (85)$$

$$\vec{p} + \vec{p}_1 = \vec{p}_2 \quad (86)$$

$$p^2 = 2m^2 - 2\vec{p}_1 \cdot (\vec{p} + \vec{p}_1) = 2m^2 - 2p_1^2 - 2\vec{p} \cdot \vec{p}_1 = -2\vec{p} \cdot \vec{p}_1 \\ \implies 2\vec{p} \cdot \vec{p}_1 = -p^2 \quad (87)$$

$$D = (\vec{k} + y\vec{p} - z\vec{p}_1)^2 + y(1-y-z)p^2 - (1-z)^2 m^2 - z\lambda^2 + i\epsilon \quad (88)$$

$$D = (\vec{k} + y\vec{p} - z\vec{p}_1)^2 + xyp^2 - (1-z)^2 m^2 - z\lambda^2 + i\epsilon \quad (89)$$

$$\Delta = (1-z)^2 m^2 - xyp^2 + z\lambda^2 \quad (90)$$

$$D = (\vec{k} + y\vec{p} - z\vec{p}_1)^2 - \Delta + i\epsilon \quad (91)$$

$$\star = -\frac{e^3}{(2\pi)^4} \int d^4 k \frac{\bar{u}_2 \gamma^\nu (\not{p} + \not{k} + m) \gamma^\mu (\not{k} + m) \gamma_\nu u_1}{ABC} \quad (92)$$

$$\frac{1}{ABC} = 2 \int_0^1 dx dy dz \frac{\delta(x+y+z-1)}{D^3} \quad (93)$$

$$\star = -\frac{2e^3}{(2\pi)^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4 k \frac{\bar{u}_2 \gamma^\nu (\not{p} + \not{k} + m) \gamma^\mu (\not{k} + m) \gamma_\nu u_1}{D^3} \quad (94)$$

$$\star = -\frac{2e^3}{(2\pi)^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4 k \frac{\bar{u}_2 \gamma^\nu (\not{p} + \not{k} + m) \gamma^\mu (\not{k} + m) \gamma_\nu u_1}{[(\vec{k} + y\vec{p} - z\vec{p}_1)^2 - \Delta + i\epsilon]^3} \quad (95)$$

$$\star = -\frac{2e^3}{(2\pi)^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4 k \frac{N}{[(\vec{k} + y\vec{p} - z\vec{p}_1)^2 - \Delta + i\epsilon]^3} \quad (96)$$

$$N = \bar{u}_2 \gamma^\nu (\not{p} + \not{k} + m) \gamma^\mu (\not{k} + m) \gamma_\nu u_1 \quad (97)$$

$$N = \bar{u}_2 (\gamma^\nu \not{p} + \gamma^\nu \not{k} + m \gamma^\nu) \gamma^\mu (\not{k} + m) \gamma_\nu u_1 \quad (98)$$

$$\gamma^\nu \not{p} = \gamma^\nu \gamma^\beta p_\beta \quad (99)$$

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} I \implies \gamma^\mu \gamma^\nu = 2\eta^{\mu\nu} I - \gamma^\nu \gamma^\mu \quad (100)$$

Where I is the 4×4 identity matrix.

$$\begin{aligned} \gamma^\nu \not{p} &= (2\eta^{\nu\beta} I - \gamma^\beta \gamma^\nu) p_\beta = 2\eta^{\nu\beta} p_\beta I - \gamma^\beta p_\beta \gamma^\nu \\ &= 2p^\nu I - \not{p} \gamma^\nu = 2p^\nu - \not{p} \gamma^\nu \end{aligned} \quad (101)$$

In the last step I suppress I to condense the math.

$$N = \bar{u}_2 [(m - \not{p} - \not{k}) \gamma^\nu + 2(p^\nu + k^\nu)] \gamma^\mu (\not{k} + m) \gamma_\nu u_1 \quad (102)$$

$$N = \bar{u}_2 [(m - \not{p} - \not{k}) \gamma^\nu \gamma^\mu + 2(p^\nu + k^\nu) \gamma^\mu] (\not{k} + m) \gamma_\nu u_1 \quad (103)$$

$$\begin{aligned} N &= \bar{u}_2 [(m - \not{p} - \not{k}) (2\eta^{\nu\mu} I - \gamma^\mu \gamma^\nu) + 2\gamma^\mu (p^\nu + k^\nu)] \\ &\quad (\not{k} + m) \gamma_\nu u_1 \end{aligned} \quad (104)$$

$$\begin{aligned} N &= \bar{u}_2 [(m - \not{p} - \not{k}) (2\eta^{\nu\mu} - \gamma^\mu \gamma^\nu) + 2\gamma^\mu (p^\nu + k^\nu)] \\ &\quad (\not{k} + m) \gamma_\nu u_1 \end{aligned} \quad (105)$$

$$\begin{aligned} N &= \bar{u}_2 [2(m - \not{p} - \not{k}) \eta^{\nu\mu} - (m - \not{p} - \not{k}) \gamma^\mu \gamma^\nu + 2\gamma^\mu (p^\nu + k^\nu)] \\ &\quad (\not{k} + m) \gamma_\nu u_1 \end{aligned} \quad (106)$$

$$\begin{aligned} N &= \bar{u}_2 [2(m - \not{p} - \not{k}) (\not{k} + m) \eta^{\nu\mu} \\ &\quad - (m - \not{p} - \not{k}) \gamma^\mu \gamma^\nu (\not{k} + m) + 2\gamma^\mu (\not{k} + m) (p^\nu + k^\nu)] \gamma_\nu u_1 \end{aligned} \quad (107)$$

$$\begin{aligned} N &= \bar{u}_2 [2(m - \not{p} - \not{k}) (\not{k} + m) \eta^{\nu\mu} \\ &\quad - (m - \not{p} - \not{k}) \gamma^\mu (\gamma^\nu \not{k} + m \gamma^\nu) + 2\gamma^\mu (\not{k} + m) (p^\nu + k^\nu)] \gamma_\nu u_1 \end{aligned} \quad (108)$$

$$\begin{aligned} N &= \bar{u}_2 [2(m - \not{p} - \not{k}) (\not{k} + m) \eta^{\nu\mu} + 2\gamma^\mu (\not{k} + m) (p^\nu + k^\nu) \\ &\quad - (m - \not{p} - \not{k}) \gamma^\mu (2k^\nu - \not{k} \gamma^\nu + m \gamma^\nu)] \gamma_\nu u_1 \end{aligned} \quad (109)$$

$$\begin{aligned} N &= \bar{u}_2 [2(m - \not{p} - \not{k}) (\not{k} + m) \eta^{\nu\mu} + 2\gamma^\mu (\not{k} + m) (p^\nu + k^\nu) \\ &\quad - 2(m - \not{p} - \not{k}) \gamma^\mu k^\nu + (m - \not{p} - \not{k}) \gamma^\mu (\not{k} - m) \gamma^\nu] \gamma_\nu u_1 \end{aligned} \quad (110)$$

$$\begin{aligned} N &= \bar{u}_2 [2(m - \not{p} - \not{k}) (\not{k} + m) \eta^{\nu\mu} \gamma_\nu \\ &\quad + 2\gamma^\mu (\not{k} + m) (p^\nu + k^\nu) \gamma_\nu - 2(m - \not{p} - \not{k}) \gamma^\mu k^\nu \gamma_\nu \\ &\quad + (m - \not{p} - \not{k}) \gamma^\mu (\not{k} - m) \gamma^\nu \gamma_\nu] u_1 \end{aligned} \quad (111)$$

$$\gamma^\nu \gamma_\nu = (\gamma^0)^2 - (\gamma^1)^2 - (\gamma^2)^2 - (\gamma^3)^2 \quad (112)$$

From my QED notes

$$(\gamma^0)^2 = -(\gamma^a)^2 = I. \quad (113)$$

$$\gamma^\nu \gamma_\nu = 4I \quad (114)$$

$$\begin{aligned} N &= \bar{u}_2 [2(m - \not{p} - \not{k}) (\not{k} + m) \gamma^\mu + 2\gamma^\mu (\not{k} + m) (\not{p} + \not{k}) \\ &\quad - 2(m - \not{p} - \not{k}) \gamma^\mu \not{k} + 4(m - \not{p} - \not{k}) \gamma^\mu (\not{k} - m) I] u_1 \end{aligned} \quad (115)$$

$$\begin{aligned} N &= 2\bar{u}_2 [(m - \not{p} - \not{k}) (\not{k} + m) \gamma^\mu + \gamma^\mu (\not{k} + m) (\not{p} + \not{k}) \\ &\quad - (m - \not{p} - \not{k}) \gamma^\mu \not{k} + 2(m - \not{p} - \not{k}) \gamma^\mu (\not{k} - m)] u_1 \end{aligned} \quad (116)$$

$$\begin{aligned} N &= 2\bar{u}_2 [(m - \not{p} - \not{k}) (\not{k} \gamma^\mu + m \gamma^\mu) + (\gamma^\mu \not{k} + m \gamma^\mu) (\not{p} + \not{k}) \\ &\quad - m \gamma^\mu \not{k} + \not{p} \gamma^\mu \not{k} + \not{k} \gamma^\mu \not{k} + 2(m \gamma^\mu - \not{p} \gamma^\mu - \not{k} \gamma^\mu) (\not{k} - m)] u_1 \end{aligned} \quad (117)$$

$$\begin{aligned} N &= 2\bar{u}_2 [(m - \not{p} - \not{k}) \not{k} \gamma^\mu + (m - \not{p} - \not{k}) m \gamma^\mu + \gamma^\mu \not{k} (\not{p} + \not{k}) \\ &\quad + m \gamma^\mu (\not{p} + \not{k}) - m \gamma^\mu \not{k} + \not{p} \gamma^\mu \not{k} + \not{k} \gamma^\mu \not{k} + 2m \gamma^\mu (\not{k} - m) \\ &\quad - 2\not{p} \gamma^\mu (\not{k} - m) - 2\not{k} \gamma^\mu (\not{k} - m)] u_1 \end{aligned} \quad (118)$$

$$\begin{aligned} N &= 2\bar{u}_2 [m (\not{k} \gamma^\mu - \not{p} \gamma^\mu - \not{k} \gamma^\mu + \gamma^\mu \not{p} + \gamma^\mu \not{k} - \gamma^\mu \not{k}) \\ &\quad + 2\gamma^\mu \not{k} + 2\not{p} \gamma^\mu + 2\not{k} \gamma^\mu - \not{p} \not{k} \gamma^\mu - \not{k}^2 \gamma^\mu + \gamma^\mu \not{k} \not{p} + \gamma^\mu \not{k}^2 \\ &\quad + \not{p} \gamma^\mu \not{k} + \not{k} \gamma^\mu \not{k} - 2\not{p} \gamma^\mu \not{k} - 2\not{k} \gamma^\mu \not{k} - m^2 \gamma^\mu] u_1 \end{aligned} \quad (119)$$

$$\begin{aligned} N &= 2\bar{u}_2 [m (2\not{k} \gamma^\mu + 2\gamma^\mu \not{k} + \gamma^\mu \not{p} + \not{p} \gamma^\mu) - \not{p} \not{k} \gamma^\mu - \not{k}^2 \gamma^\mu \\ &\quad + \gamma^\mu \not{k} \not{p} + \gamma^\mu \not{k}^2 - \not{p} \gamma^\mu \not{k} - \not{k} \gamma^\mu \not{k} - m^2 \gamma^\mu] u_1 \end{aligned} \quad (120)$$

$$\begin{aligned} N &= 2\bar{u}_2 [2m (2k^\mu + p^\mu) - \not{p} \not{k} \gamma^\mu - \not{k}^2 \gamma^\mu \\ &\quad + \gamma^\mu \not{k} \not{p} + \gamma^\mu \not{k}^2 - \not{p} \gamma^\mu \not{k} - \not{k} \gamma^\mu \not{k} - m^2 \gamma^\mu] u_1 \end{aligned} \quad (121)$$

$$\begin{aligned} \not{p} \gamma^\mu &= p_\beta \gamma^\beta \gamma^\mu = p_\beta (2\eta^{\beta\mu} - \gamma^\mu \gamma^\beta) = 2\eta^{\beta\mu} p_\beta - \gamma^\mu p_\beta \gamma^\beta \\ &= 2p^\mu - \gamma^\mu \not{p} \end{aligned} \quad (122)$$

$$\not{k}^2 \gamma^\mu = \not{k} \not{k} \gamma^\mu = \not{k} (2k^\mu - \gamma^\mu \not{k}) = 2\not{k} k^\mu - \not{k} \gamma^\mu \not{k} \quad (123)$$

$$\begin{aligned} N &= 2\bar{u}_2 [2m (2k^\mu + p^\mu) - \not{p} \not{k} \gamma^\mu - 2\not{k} k^\mu \\ &\quad + \gamma^\mu \not{k} \not{p} + \gamma^\mu \not{k}^2 - \not{p} \gamma^\mu \not{k} - m^2 \gamma^\mu] u_1 \end{aligned} \quad (124)$$

$$\begin{aligned} N &= 2\bar{u}_2 [2m (2k^\mu + p^\mu) - \not{p} \not{k} \gamma^\mu - 2\not{k} k^\mu \\ &\quad + \gamma^\mu \not{k} \not{p} + \gamma^\mu \not{k}^2 - \not{p} \gamma^\mu \not{k} - m^2 \gamma^\mu] u_1 \end{aligned} \quad (125)$$

$$\gamma^\mu k^2 = \gamma^\mu k k = (2k^\mu - k\gamma^\mu)k = 2kk^\mu - k\gamma^\mu k \quad (126)$$

$$N = 2\bar{u}_2[2m(2k^\mu + p^\mu) - \not{p}k\gamma^\mu + \gamma^\mu k\not{p} - k\gamma^\mu k - \not{p}\gamma^\mu k - m^2\gamma^\mu]u_1 \quad (127)$$

$$N = 2\bar{u}_2[2m(2k^\mu + p^\mu) + \gamma^\mu k\not{p} - k\gamma^\mu k - 2\not{p}k^\mu - m^2\gamma^\mu]u_1 \quad (128)$$

$$\gamma^\mu k\not{p} = (2k^\mu - k\gamma^\mu)\not{p} = 2\not{p}k^\mu - k\gamma^\mu \not{p} \quad (129)$$

$$N = 2\bar{u}_2[2m(2k^\mu + p^\mu) - k\gamma^\mu \not{p} - k\gamma^\mu k - m^2\gamma^\mu]u_1 \quad (130)$$

$$\begin{aligned} \star &= -\frac{2e^3}{(2\pi)^4} \int_0^1 dx dy dz \delta(x+y+z-1) \\ &\int d^4k \frac{2\bar{u}_2[2m(2k^\mu + p^\mu) - k\gamma^\mu \not{p} - k\gamma^\mu k - m^2\gamma^\mu]u_1}{[(\vec{k} + y\vec{p} - z\vec{p}_1)^2 - \Delta + i\epsilon]^3} \end{aligned} \quad (131)$$

$$\begin{aligned} \star &= \frac{4e^3}{(2\pi)^4} \int_0^1 dx dy dz \delta(x+y+z-1) \\ &\int d^4k \frac{\bar{u}_2[k\gamma^\mu \not{p} + k\gamma^\mu k + m^2\gamma^\mu - 2m(2k^\mu + p^\mu)]u_1}{[(\vec{k} + y\vec{p} - z\vec{p}_1)^2 - \Delta + i\epsilon]^3} \end{aligned} \quad (132)$$

I shift the integrand.

$$\vec{q} = \vec{k} + y\vec{p} - z\vec{p}_1, \quad d^4k = d^4q, \quad \vec{k} = \vec{q} - y\vec{p} + z\vec{p}_1 \quad (133)$$

$$\star = \frac{4e^3}{(2\pi)^4} \int_0^1 dx dy dz \int d^4q \frac{\delta(x+y+z-1)\mathcal{N}}{(q^2 - \Delta + i\epsilon)^3} \quad (134)$$

$$\begin{aligned} \mathcal{N} &= \bar{u}_2[(\not{q} - y\not{p} + z\not{p}_1)\gamma^\mu \not{p} \\ &+ (\not{q} - y\not{p} + z\not{p}_1)\gamma^\mu (\not{q} - y\not{p} + z\not{p}_1) + m^2\gamma^\mu \\ &- 2m(2q^\mu - 2yp^\mu + 2zp_1^\mu + p^\mu)]u_1 \end{aligned} \quad (135)$$

$$\begin{aligned} \mathcal{N} &= \bar{u}_2[(1-y)\not{q}\gamma^\mu \not{p} - y\not{p}\gamma^\mu \not{q} + z\not{q}\gamma^\mu \not{p}_1 + z\not{p}_1\gamma^\mu \not{q} - 4mq^\mu \\ &+ z(1-y)\not{p}_1\gamma^\mu \not{p} - z\not{p}\gamma^\mu \not{p}_1 + y(y-1)\not{p}\gamma^\mu \not{p} + \not{q}\gamma^\mu \not{q} + z^2\not{p}_1\gamma^\mu \not{p}_1 \\ &+ m^2\gamma^\mu - 2m(2zp_1^\mu + p^\mu - 2yp^\mu)]u_1 \end{aligned} \quad (136)$$

$$\mathcal{N} = \bar{u}_2[(1-y)\not{q}\gamma^\mu \not{p} - y\not{p}\gamma^\mu \not{q} + z\not{q}\gamma^\mu \not{p}_1 + z\not{p}_1\gamma^\mu \not{q} - 4mq^\mu]u_1 + \mathcal{M} \quad (137)$$

$$\begin{aligned} \mathcal{M} &= \bar{u}_2[z(1-y)\not{p}_1\gamma^\mu \not{p} - zy\not{p}\gamma^\mu \not{p}_1 + y(y-1)\not{p}\gamma^\mu \not{p} + \not{q}\gamma^\mu \not{q} \\ &+ z^2\not{p}_1\gamma^\mu \not{p}_1 + m^2\gamma^\mu - 2m(2zp_1^\mu + p^\mu - 2yp^\mu)]u_1 \end{aligned} \quad (138)$$

$$\begin{aligned} \star &= \frac{4e^3}{(2\pi)^4} \int_0^1 dx dy dz \int d^4q \frac{\delta(x+y+z-1)}{(q^2 - \Delta + i\epsilon)^3} [\mathcal{M} + \\ &\bar{u}_2[(1-y)\not{q}\gamma^\mu \not{p} - y\not{p}\gamma^\mu \not{q} + z\not{q}\gamma^\mu \not{p}_1 + z\not{p}_1\gamma^\mu \not{q} - 4mq^\mu]u_1] \end{aligned} \quad (139)$$

Terms with one factor of q are odd and integrate to 0.

$$\star = \frac{4e^3}{(2\pi)^4} \int_0^1 dx dy dz \int d^4q \frac{\delta(x+y+z-1)}{(q^2 - \Delta + i\epsilon)^3} \mathcal{M} \quad (140)$$

$$\mathcal{M} = \bar{u}_2(\not{q}\gamma^\mu \not{q} + \mathcal{N})u_1 = \bar{u}_2(q^\alpha q^\beta \gamma_\alpha \gamma^\mu \gamma_\beta + \mathcal{N})u_1 \quad (141)$$

$$\begin{aligned} \mathcal{N} &= z(1-y)\not{p}_1\gamma^\mu \not{p} - zy\not{p}\gamma^\mu \not{p}_1 + y(y-1)\not{p}\gamma^\mu \not{p} \\ &+ z^2\not{p}_1\gamma^\mu \not{p}_1 + m^2\gamma^\mu - 2m(2zp_1^\mu + p^\mu - 2yp^\mu) \end{aligned} \quad (142)$$

$$\begin{aligned} \star &= \frac{4e^3}{(2\pi)^4} \int_0^1 dx dy dz \delta(x+y+z-1) \\ &\int d^4q \frac{\bar{u}_2(q^\alpha q^\beta \gamma_\alpha \gamma^\mu \gamma_\beta + \mathcal{N})u_1}{(q^2 - \Delta + i\epsilon)^3} \end{aligned} \quad (143)$$

$$\begin{aligned} \star &= \frac{4e^3}{(2\pi)^4} \int_0^1 dx dy dz \delta(x+y+z-1) \\ &\int d^4q \frac{\bar{u}_2 \mathcal{N} u_1}{(q^2 - \Delta + i\epsilon)^3} + 4e\odot \end{aligned} \quad (144)$$

$$\begin{aligned} \odot &= \frac{e^2}{(2\pi)^4} \int_0^1 dx dy dz \delta(x+y+z-1) \\ &\int d^4q \frac{\bar{u}_2 q^\alpha q^\beta \gamma_\alpha \gamma^\mu \gamma_\beta u_1}{(q^2 - \Delta + i\epsilon)^3} \end{aligned} \quad (145)$$

$$\star = \frac{2e^3}{(2\pi)^4} \int_0^1 dx dy dz \int d^4q \frac{\delta(x+y+z-1)}{(q^2 - \Delta + i\epsilon)^3} \mathbb{N} + 4e\odot \quad (146)$$

$$\mathbb{N} = 2\bar{u}_2 \mathcal{N} u_1 \quad (147)$$

The Gordon identity, which holds for on-shell spinors, is

$$\begin{aligned} \bar{u}(\vec{p})(q^\mu + p^\mu)u(\vec{q}) &= \\ &2m\bar{u}(\vec{p})\gamma^\mu u(\vec{q}) + i\bar{u}(\vec{p})\sigma^{\mu\nu}(q_\nu - p_\nu)u(\vec{q}), \end{aligned} \quad (148)$$

where

$$\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]. \quad (149)$$

$$\begin{aligned} \mathcal{N} &= z(1-y)\not{p}_1\gamma^\mu \not{p} - zy\not{p}\gamma^\mu \not{p}_1 + y(y-1)\not{p}\gamma^\mu \not{p} \\ &+ z^2\not{p}_1\gamma^\mu \not{p}_1 + m^2\gamma^\mu - 2m(2zp_1^\mu + p^\mu - 2yp^\mu) \end{aligned} \quad (150)$$

$$\not{p}\gamma^\mu = 2p^\mu - \gamma^\mu \not{p} \quad (151)$$

$$\not{p}_1 \gamma^\mu \not{p} = (2p_1^\mu - \gamma^\mu \not{p}_1) \not{p} = 2p_1^\mu \not{p} - \gamma^\mu \not{p}_1 \not{p} \quad (152)$$

$$\not{q}\not{p} = q_\alpha p_\beta \gamma^\alpha \gamma^\beta = q_\alpha p_\beta (2\eta^{\alpha\beta} - \gamma^\beta \gamma^\alpha) = 2\eta^{\alpha\beta} q_\alpha p_\beta - q_\alpha p_\beta \gamma^\beta \gamma^\alpha = 2\vec{q} \cdot \vec{p} - \not{p}\not{q} \quad (153)$$

$$\not{p}_1 \gamma^\mu \not{p} = 2p_1^\mu \not{p} - 2\gamma^\mu \vec{p}_1 \cdot \vec{p} + \gamma^\mu \not{p} \not{p}_1 \quad (154)$$

I use Eq. 87.

$$2\vec{p} \cdot \vec{p}_1 = -p^2 \quad (155)$$

$$\not{p}_1 \gamma^\mu \not{p} = 2p_1^\mu \not{p} + p^2 \gamma^\mu + \gamma^\mu \not{p} \not{p}_1 \quad (156)$$

$$\gamma^\mu \not{p} = 2p^\mu - \not{p}\gamma^\mu \quad (157)$$

$$\not{p}_1 \gamma^\mu \not{p} = 2p_1^\mu \not{p} + p^2 \gamma^\mu + 2p_1^\mu \not{p}_1 - \not{p}\gamma^\mu \not{p}_1 \quad (158)$$

$$\mathcal{N} = (1-y)(2zp_1^\mu - y\not{p}\gamma^\mu) \not{p} + z[(1-y)2p^\mu - \not{p}\gamma^\mu + z\not{p}_1 \gamma^\mu] \not{p}_1 + [m^2 + z(1-y)p^2] \gamma^\mu - 2m(2zp_1^\mu + p^\mu - 2yp^\mu) \quad (159)$$

$$\mathbb{N} = 2\bar{u}_2[(1-y)(2zp_1^\mu - y\not{p}\gamma^\mu) \not{p} + [m^2 + z(1-y)p^2] \gamma^\mu + z[(1-y)2p^\mu - \not{p}\gamma^\mu + z\not{p}_1 \gamma^\mu] \not{p}_1 - 2m(2zp_1^\mu + p^\mu - 2yp^\mu)] u_1 \quad (160)$$

$$\mathbb{N} = 2(1-y)\bar{u}_2(2zp_1^\mu - y\not{p}\gamma^\mu) \not{p} u_1 + 2\bar{u}_2 z[(1-y)2p^\mu - \not{p}\gamma^\mu + z\not{p}_1 \gamma^\mu] \not{p}_1 u_1 - 4m(2zp_1^\mu + p^\mu - 2yp^\mu) \bar{u}_2 u_1 + [2m^2 + 2z(1-y)p^2] \bar{u}_2 \gamma^\mu u_1 \quad (161)$$

On-shell spinors satisfy

$$\not{p} u(\vec{p}) = m u(\vec{p}), \quad \bar{u}(\vec{p}) \not{p} = m \bar{u}(\vec{p}). \quad (162)$$

$$\mathbb{N} = 2(1-y)\bar{u}_2(2zp_1^\mu - y\not{p}\gamma^\mu) \not{p} u_1 + 2mz\bar{u}_2[2(1-y)p^\mu - \not{p}\gamma^\mu + z\not{p}_1 \gamma^\mu] u_1 - 4m(2zp_1^\mu + p^\mu - 2yp^\mu) \bar{u}_2 u_1 + [2m^2 + 2z(1-y)p^2] \bar{u}_2 \gamma^\mu u_1 \quad (163)$$

$$\mathbb{N} = 2(1-y)\bar{u}_2(2zp_1^\mu - y\not{p}\gamma^\mu) \not{p} u_1 + 2mz\bar{u}_2[2(1-y)p^\mu - \not{p}\gamma^\mu + 2zp_1^\mu - z\gamma^\mu \not{p}_1] u_1 - 4m(2zp_1^\mu + p^\mu - 2yp^\mu) \bar{u}_2 u_1 + [2m^2 + 2z(1-y)p^2] \bar{u}_2 \gamma^\mu u_1 \quad (164)$$

$$\mathbb{N} = 2(1-y)\bar{u}_2(2zp_1^\mu - y\not{p}\gamma^\mu) \not{p} u_1 - 2mz\bar{u}_2 \not{p}\gamma^\mu u_1 + 4m[(z-yz-1+2y)p^\mu + z(z-2)p_1^\mu] \bar{u}_2 u_1 + [2m^2 + 2z(1-y)p^2] \bar{u}_2 \gamma^\mu u_1 - 2mz^2 \bar{u}_2 \gamma^\mu \not{p}_1 u_1 \quad (165)$$

$$\mathbb{N} = 2(1-y)\bar{u}_2(2zp_1^\mu - y\not{p}\gamma^\mu) \not{p} u_1 - 2mz\bar{u}_2 \not{p}\gamma^\mu u_1 + 4m[(z-yz-1+2y)p^\mu + z(z-2)p_1^\mu] \bar{u}_2 u_1 + [2m^2 + 2z(1-y)p^2] \bar{u}_2 \gamma^\mu u_1 - 2m^2 z^2 \bar{u}_2 \gamma^\mu u_1 \quad (166)$$

$$\mathbb{N} = 2(1-y)\bar{u}_2(2zp_1^\mu - y\not{p}\gamma^\mu) \not{p} u_1 - 2mz\bar{u}_2 \not{p}\gamma^\mu u_1 + 4m[(z-yz-1+2y)p^\mu + z(z-2)p_1^\mu] \bar{u}_2 u_1 + [2m^2(1-z^2) + 2z(1-y)p^2] \bar{u}_2 \gamma^\mu u_1 \quad (167)$$

I use Eq. 67

$$\not{p} = \not{p}_2 - \not{p}_1. \quad (168)$$

$$\mathbb{N} = 2(1-y)\bar{u}_2(2zp_1^\mu - y\not{p}_2 \gamma^\mu + y\not{p}_1 \gamma^\mu) (\not{p}_2 - \not{p}_1) u_1 + 4m[(z-yz-1+2y)p^\mu + z(z-2)p_1^\mu] \bar{u}_2 u_1 + [2m^2(1-z^2) + 2z(1-y)p^2] \bar{u}_2 \gamma^\mu u_1 - 2mz\bar{u}_2 (\not{p}_2 - \not{p}_1) \gamma^\mu u_1 \quad (169)$$

$$\mathbb{N} = 2(1-y)(2zp_1^\mu \bar{u}_2 - y\bar{u}_2 \not{p}_2 \gamma^\mu + y\bar{u}_2 \not{p}_1 \gamma^\mu) (\not{p}_2 u_1 - \not{p}_1 u_1) + 4m[(z-yz-1+2y)p^\mu + z(z-2)p_1^\mu] \bar{u}_2 u_1 + [2m^2(1-z^2) + 2z(1-y)p^2] \bar{u}_2 \gamma^\mu u_1 - 2mz(\bar{u}_2 \not{p}_2 - \bar{u}_2 \not{p}_1) \gamma^\mu u_1 \quad (170)$$

$$\mathbb{N} = 2(1-y)(2zp_1^\mu \bar{u}_2 - my\bar{u}_2 \gamma^\mu + y\bar{u}_2 \not{p}_1 \gamma^\mu) (\not{p}_2 u_1 - mu_1) + 4m[(z-yz-1+2y)p^\mu + z(z-2)p_1^\mu] \bar{u}_2 u_1 + [2m^2(1-z^2) + 2z(1-y)p^2] \bar{u}_2 \gamma^\mu u_1 - 2mz(m\bar{u}_2 - \bar{u}_2 \not{p}_1) \gamma^\mu u_1 \quad (171)$$

$$\mathbb{N} = 4(1-y)zp_1^\mu \bar{u}_2 \not{p}_2 u_1 - 4m(1-y)zp_1^\mu \bar{u}_2 u_1 - 2(1-y)my\bar{u}_2 \gamma^\mu \not{p}_2 u_1 + 2m^2(1-y)y\bar{u}_2 \gamma^\mu u_1 + 2(1-y)y\bar{u}_2 \not{p}_1 \gamma^\mu \not{p}_2 u_1 - 2m(1-y)y\bar{u}_2 \not{p}_1 \gamma^\mu u_1 + 4m[(z-yz-1+2y)p^\mu + z(z-2)p_1^\mu] \bar{u}_2 u_1 + [2m^2(1-z^2) + 2z(1-y)p^2] \bar{u}_2 \gamma^\mu u_1 - 2m^2 z \bar{u}_2 \gamma^\mu u_1 + 2mz\bar{u}_2 \not{p}_1 \gamma^\mu u_1 \quad (172)$$

$$\mathbb{N} = 4m(1-y)zp_1^\mu \bar{u}_2 u_1 - 4m(1-y)zp_1^\mu \bar{u}_2 u_1 - 2(1-y)my\bar{u}_2 \gamma^\mu \not{p}_2 u_1 + 2m^2(1-y)y\bar{u}_2 \gamma^\mu u_1 + 2(1-y)y\bar{u}_2 \not{p}_1 \gamma^\mu \not{p}_2 u_1 - 2m(1-y)y\bar{u}_2 \not{p}_1 \gamma^\mu u_1 + 4m[(z-yz-1+2y)p^\mu + z(z-2)p_1^\mu] \bar{u}_2 u_1 + [2m^2(1-z^2) + 2z(1-y)p^2] \bar{u}_2 \gamma^\mu u_1 - 2m^2 z \bar{u}_2 \gamma^\mu u_1 + 2mz\bar{u}_2 \not{p}_1 \gamma^\mu u_1 \quad (173)$$

$$\begin{aligned}\mathbb{N} = & 2mz\bar{u}_2\cancel{p}_1\gamma^\mu u_1 - 2(1-y)m y\bar{u}_2\gamma^\mu\cancel{p}_2 u_1 \\ & + 2(1-y)y\bar{u}_2\cancel{p}_1\gamma^\mu\cancel{p}_2 u_1 - 2m(1-y)y\bar{u}_2\cancel{p}_1\gamma^\mu u_1 \\ & + 4m[(z-yz-1+2y)p^\mu + z(z-2)p_1^\mu]\bar{u}_2 u_1 \\ & + [2m^2(1+y-z-z^2-y^2) + 2z(1-y)p^2]\bar{u}_2\gamma^\mu u_1\end{aligned}\quad (174)$$

$$\begin{aligned}\mathbb{N} = & 2m(z-y+y^2)\bar{u}_2\cancel{p}_1\gamma^\mu u_1 - 2m(1-y)y\bar{u}_2\gamma^\mu\cancel{p}_2 u_1 \\ & + 2(1-y)y\bar{u}_2\cancel{p}_1\gamma^\mu\cancel{p}_2 u_1 \\ & + 4m[(z-yz-1+2y)p^\mu + z(z-2)p_1^\mu]\bar{u}_2 u_1 \\ & + [2m^2(1+y-z-z^2-y^2) + 2z(1-y)p^2]\bar{u}_2\gamma^\mu u_1\end{aligned}\quad (175)$$

$$\cancel{p}_1\gamma^\mu = 2p_1^\mu - \gamma^\mu\cancel{p}_1, \quad \gamma^\mu\cancel{p}_2 = 2p_2^\mu - \cancel{p}_2\gamma^\mu \quad (176)$$

$$\begin{aligned}\mathbb{N} = & 2m(z-y+y^2)(2p_1^\mu\bar{u}_2 u_1 - \bar{u}_2\gamma^\mu\cancel{p}_1 u_1) \\ & + 2(1-y)y\bar{u}_2\cancel{p}_1\gamma^\mu\cancel{p}_2 u_1 - 2m(1-y)y(2p_2^\mu\bar{u}_2 u_1 - \bar{u}_2\cancel{p}_2\gamma^\mu u_1) \\ & + 4m[(z-yz-1+2y)p^\mu + z(z-2)p_1^\mu]\bar{u}_2 u_1 \\ & + [2m^2(1+y-z-z^2-y^2) + 2z(1-y)p^2]\bar{u}_2\gamma^\mu u_1\end{aligned}\quad (177)$$

$$\begin{aligned}\mathbb{N} = & 2m(z-y+y^2)(2p_1^\mu\bar{u}_2 u_1 - m\bar{u}_2\gamma^\mu u_1) \\ & + 2(1-y)y\bar{u}_2\cancel{p}_1\gamma^\mu\cancel{p}_2 u_1 - 2m(1-y)y(2p_2^\mu\bar{u}_2 u_1 - m\bar{u}_2\gamma^\mu u_1) \\ & + 4m[(z-yz-1+2y)p^\mu + z(z-2)p_1^\mu]\bar{u}_2 u_1 \\ & + [2m^2(1+y-z-z^2-y^2) + 2z(1-y)p^2]\bar{u}_2\gamma^\mu u_1\end{aligned}\quad (178)$$

$$\begin{aligned}\mathbb{N} = & 2(1-y)y\bar{u}_2\cancel{p}_1\gamma^\mu\cancel{p}_2 u_1 + 4m[(z-yz-1+2y)p^\mu \\ & + (z^2-2z+z-y+y^2)p_1^\mu - y(1-y)p_2^\mu]\bar{u}_2 u_1 \\ & + [2m^2(1+3y-2z-z^2-3y^2) + 2z(1-y)p^2]\bar{u}_2\gamma^\mu u_1\end{aligned}\quad (179)$$

Eq. 67 implies

$$p_2^\mu = p^\mu + p_1^\mu. \quad (180)$$

$$\begin{aligned}\mathbb{N} = & 2(1-y)y\bar{u}_2\cancel{p}_1\gamma^\mu\cancel{p}_2 u_1 \\ & + 4m[(z-yz-1+y+y^2)p^\mu + (z^2-z-2y+2y^2)p_1^\mu]\bar{u}_2 u_1 \\ & + [2m^2(1+3y-2z-z^2-3y^2) + 2z(1-y)p^2]\bar{u}_2\gamma^\mu u_1\end{aligned}\quad (181)$$

$$\cancel{p}_1\gamma^\mu\cancel{p}_2 = (2p_1^\mu - \gamma^\mu\cancel{p}_1)\cancel{p}_2 = 2p_1^\mu\cancel{p}_2 - \gamma^\mu\cancel{p}_1\cancel{p}_2 \quad (182)$$

I use Eq. 153

$$\cancel{p}\cancel{p} = 2\vec{q} \cdot \vec{p} - \cancel{p}\cancel{q}. \quad (183)$$

$$\begin{aligned}\cancel{p}_1\gamma^\mu\cancel{p}_2 = & 2p_1^\mu\cancel{p}_2 - \gamma^\mu(2\vec{p}_1 \cdot \vec{p}_2 - \cancel{p}_2\cancel{p}_1) = \\ & 2p_1^\mu\cancel{p}_2 - 2\vec{p}_1 \cdot \vec{p}_2\gamma^\mu + \gamma^\mu\cancel{p}_2\cancel{p}_1\end{aligned}\quad (184)$$

I use Eq. 85

$$2\vec{p}_1 \cdot \vec{p}_2 = 2m^2 - p^2. \quad (185)$$

$$\begin{aligned}\cancel{p}_1\gamma^\mu\cancel{p}_2 = & 2p_1^\mu\cancel{p}_2 + (p^2 - 2m^2)\gamma^\mu + \gamma^\mu\cancel{p}_2\cancel{p}_1 = \\ & 2p_1^\mu\cancel{p}_2 + (p^2 - 2m^2)\gamma^\mu + (2p_2^\mu - \cancel{p}_2\gamma^\mu)\cancel{p}_1 = \\ & 2p_1^\mu\cancel{p}_2 + (p^2 - 2m^2)\gamma^\mu + 2p_2^\mu\cancel{p}_1 - \cancel{p}_2\gamma^\mu\cancel{p}_1 = \\ & 2p_1^\mu\cancel{p}_2 + (p^2 - 2m^2)\gamma^\mu + 2(p^\mu + p_1^\mu)\cancel{p}_1 - \cancel{p}_2\gamma^\mu\cancel{p}_1\end{aligned}\quad (186)$$

$$\begin{aligned}\mathbb{N} = & 4(1-y)y p_1^\mu\bar{u}_2\cancel{p}_2 u_1 + 2(1-y)y(p^2 - 2m^2)\bar{u}_2\gamma^\mu u_1 + \\ & 4(1-y)y(p^\mu + p_1^\mu)\bar{u}_2\cancel{p}_1 u_1 - 2(1-y)y\bar{u}_2\cancel{p}_2\gamma^\mu\cancel{p}_1 u_1 \\ & + 4m[(z-yz-1+y+y^2)p^\mu + (z^2-z-2y+2y^2)p_1^\mu]\bar{u}_2 u_1 \\ & + [2m^2(1+3y-2z-z^2-3y^2) + 2z(1-y)p^2]\bar{u}_2\gamma^\mu u_1\end{aligned}\quad (187)$$

$$\begin{aligned}\mathbb{N} = & 4m(1-y)y p_1^\mu\bar{u}_2 u_1 + 4m(1-y)y(p^\mu + p_1^\mu)\bar{u}_2 u_1 \\ & - 2m^2(1-y)y\bar{u}_2\gamma^\mu u_1 \\ & + 4m[(z-yz-1+y+y^2)p^\mu + (z^2-z-2y+2y^2)p_1^\mu]\bar{u}_2 u_1 \\ & + [2m^2(1+y-2z-z^2-y^2) + 2(z-yz+y-y^2)p^2]\bar{u}_2\gamma^\mu u_1\end{aligned}\quad (188)$$

$$\begin{aligned}\mathbb{N} = & 4m[(z-yz-1+2y)p^\mu + z(z-1)p_1^\mu]\bar{u}_2 u_1 \\ & + [2m^2(1-2z-z^2) + 2(z-yz+y-y^2)p^2]\bar{u}_2\gamma^\mu u_1\end{aligned}\quad (189)$$

Eq. 67 implies

$$2\vec{p}_1 = \vec{p}_1 + \vec{p}_2 - \vec{p}. \quad (190)$$

$$\begin{aligned}\mathbb{N} = & 2m[2(z-yz-1+2y)p^\mu + z(z-1)(p_1^\mu + p_2^\mu - p^\mu)]\bar{u}_2 u_1 \\ & + [2m^2(1-2z-z^2) + 2(z-yz+y-y^2)p^2]\bar{u}_2\gamma^\mu u_1\end{aligned}\quad (191)$$

$$\begin{aligned}\mathbb{N} = & 2m[(3z-2yz-2+4y-z^2)p^\mu + z(z-1)(p_1^\mu + p_2^\mu)]\bar{u}_2 u_1 \\ & + [2m^2(1-2z-z^2) + 2(z-yz+y-y^2)p^2]\bar{u}_2\gamma^\mu u_1\end{aligned}\quad (192)$$

Using the identity $x + y + z = 1$, which holds under the integral sign, I find

$$\begin{aligned}3z-2yz-2+4y-z^2 = & (z-2)(1-2y-z) = (z-2)(x-y), \\ z-yz+y-y^2 = & (1-y)(y+z) = (1-y)(1-x).\end{aligned}\quad (193)$$

$$\mathbb{N} = 2m[(z-2)(x-y)p^\mu + z(z-1)(p_1^\mu + p_2^\mu)]\bar{u}_2 u_1 + [2m^2(1-2z-z^2) + 2(1-y)(1-x)p^2]\bar{u}_2 \gamma^\mu u_1 \quad (194)$$

$$\mathbb{N} = 2m(z-2)(x-y)p^\mu \bar{u}_2 u_1 + 2mz(z-1)\bar{u}_2(p_1^\mu + p_2^\mu)u_1 + [2m^2(1-2z-z^2) + 2(1-y)(1-x)p^2]\bar{u}_2 \gamma^\mu u_1 \quad (195)$$

I use the Gordon identity (Eq. 148).

$$\bar{u}_2(p_1^\mu + p_2^\mu)u_1 = 2m\bar{u}_2 \gamma^\mu u_1 + i\bar{u}_2 \sigma_{\mu\nu}(p_1^\nu - p_2^\nu)u_1 \quad (196)$$

$$\vec{p} = \vec{p}_2 - \vec{p}_1 \implies p_1^\nu - p_2^\nu = -p^\nu \quad (197)$$

$$\begin{aligned} \bar{u}_2(p_1^\mu + p_2^\mu)u_1 &= 2m\bar{u}_2 \gamma^\mu u_1 - i\bar{u}_2 \sigma_{\mu\nu} p^\nu u_1 \\ &= 2m\bar{u}_2 \gamma^\mu u_1 - i\bar{u}_2 \sigma^{\mu\nu} p_\nu u_1 \end{aligned} \quad (198)$$

$$\begin{aligned} \mathbb{N} &= 2m(z-2)(x-y)p^\mu \bar{u}_2 u_1 - 2miz(z-1)\bar{u}_2 \sigma^{\mu\nu} p_\nu u_1 \\ &\quad + [2m^2(1-2z-z^2) + 2(1-y)(1-x)p^2]\bar{u}_2 \gamma^\mu u_1 \\ &\quad + 4m^2 z(z-1)\bar{u}_2 \gamma^\mu u_1 \end{aligned} \quad (199)$$

$$\mathbb{N} = 2m(z-2)(x-y)p^\mu \bar{u}_2 u_1 - 2miz(z-1)p_\nu \bar{u}_2 \sigma^{\mu\nu} u_1 + [2m^2(1-4z+z^2) + 2(1-y)(1-x)p^2]\bar{u}_2 \gamma^\mu u_1 \quad (200)$$

$$\mathbb{M} = [2m^2(1-4z+z^2) + 2(1-y)(1-x)p^2]\bar{u}_2 \gamma^\mu u_1 - 2miz(z-1)p_\nu \bar{u}_2 \sigma^{\mu\nu} u_1 \quad (201)$$

$$\mathbb{N} = 2m(z-2)(x-y)p^\mu \bar{u}_2 u_1 + \mathbb{M} \quad (202)$$

$$\star = \frac{2e^3}{(2\pi)^4} \int_0^1 dx dy dz \int d^4 q \frac{\delta(x+y+z-1)}{(q^2 - \Delta + i\epsilon)^3} \mathbb{N} + 4e\odot \quad (203)$$

$$\begin{aligned} \star &= 4e\odot + \frac{2e^3}{(2\pi)^4} \left[\int_0^1 dx dy dz \int d^4 q \frac{\delta(x+y+z-1)}{(q^2 - \Delta + i\epsilon)^3} \mathbb{M} \right. \\ &\quad \left. + 2mp^\mu \bar{u}_2 u_1 \int_0^1 dx dy dz \int d^4 q \frac{\delta(x+y+z-1)}{(q^2 - \Delta + i\epsilon)^3} (z-2)(x-y) \right] \end{aligned} \quad (204)$$

$$\aleph = \int_0^1 dx dy dz \int d^4 q \frac{\delta(x+y+z-1)}{(q^2 - \Delta + i\epsilon)^3} (z-2)(x-y) \quad (205)$$

$$\begin{aligned} \star &= \frac{2e^3}{(2\pi)^4} \left[\int_0^1 dx dy dz \int d^4 q \frac{\delta(x+y+z-1)}{(q^2 - \Delta + i\epsilon)^3} \mathbb{M} \right. \\ &\quad \left. + 2mp^\mu \bar{u}_2 u_1 \aleph \right] + 4e\odot \end{aligned} \quad (206)$$

$$\aleph = \int_0^1 dx dy dz \int d^4 q \frac{\delta(x+y+z-1)}{(q^2 - \Delta + i\epsilon)^3} (z-2)(x-y) \quad (207)$$

$$\Delta = (1-z)^2 m^2 - xyp^2 + z\lambda^2 \quad (208)$$

$$\begin{aligned} \aleph &= \int_0^1 dx dy dz \\ &\quad \int d^4 q \frac{\delta(x+y+z-1)(z-2)(x-y)}{[q^2 - (1-z)^2 m^2 + xyp^2 - z\lambda^2 + i\epsilon]^3} \end{aligned} \quad (209)$$

$$h = z-2, \quad w = q^2 - (1-z)^2 m^2 - z\lambda^2 + i\epsilon \quad (210)$$

$$\aleph = \int_0^1 dx dy dz \int d^4 q \frac{h(x-y)}{(w + xyp^2)^3} \delta(x+y+z-1) \quad (211)$$

I swap dummy variables $x \leftrightarrow y$ with substitution.

$$\aleph = \int_0^1 dy dx dz \int d^4 q \frac{h(y-x)}{(w + yxp^2)^3} \delta(y+x+z-1) \quad (212)$$

$$\aleph = - \int_0^1 dx dy dz \int d^4 q \frac{h(x-y)}{(w + xyp^2)^3} \delta(x+y+z-1) \quad (213)$$

$$\aleph = -\aleph \implies \aleph = 0 \quad (214)$$

$$\begin{aligned} \star &= \frac{2e^3}{(2\pi)^4} \left[\int_0^1 dx dy dz \int d^4 q \frac{\delta(x+y+z-1)}{(q^2 - \Delta + i\epsilon)^3} \mathbb{M} \right. \\ &\quad \left. + 2mp^\mu \bar{u}_2 u_1 \aleph \right] + 4e\odot \end{aligned} \quad (215)$$

$$\star = \frac{2e^3}{(2\pi)^4} \int_0^1 dx dy dz \int d^4 q \frac{\delta(x+y+z-1)}{(q^2 - \Delta + i\epsilon)^3} \mathbb{M} + 4e\odot \quad (216)$$

$$\begin{aligned} \mathbb{M} &= 2[m^2(1-4z+z^2) + (1-y)(1-x)p^2]\bar{u}_2 \gamma^\mu u_1 \\ &\quad - 2miz(z-1)p_\nu \bar{u}_2 \sigma^{\mu\nu} u_1 \end{aligned} \quad (217)$$

$$\mathbb{J} = 2[m^2(1-4z+z^2) + (1-y)(1-x)p^2]\bar{u}_2 \gamma^\mu u_1 \quad (218)$$

$$\mathbb{M} = \mathbb{J} - 2miz(z-1)p_\nu \bar{u}_2 \sigma^{\mu\nu} u_1 \quad (219)$$

$$\begin{aligned} \star &= \frac{2e^3}{(2\pi)^4} \int_0^1 dx dy dz \int d^4 q \frac{\delta(x+y+z-1)}{(q^2 - \Delta + i\epsilon)^3} \mathbb{J} + 4e\odot \\ &\quad - \frac{4ime^3}{(2\pi)^4} p_\nu \bar{u}_2 \sigma^{\mu\nu} u_1 \int_0^1 dx dy dz \delta(x+y+z-1) z(z-1) \aleph \end{aligned} \quad (220)$$

$$\mathbb{X} = \int \frac{d^4 q}{(q^2 - \Delta + i\epsilon)^3} \quad (221)$$

$$\begin{aligned} \star &= 4e\odot + \frac{4e^3}{(2\pi)^4} \bar{u}_2 \gamma^\mu u_1 \int_0^1 dx dy dz \delta(x+y+z-1) \\ &\quad [m^2(1-4z+z^2) + (1-y)(1-x)p^2] \mathbb{X} \\ &\quad - \frac{4ime^3}{(2\pi)^4} p_\nu \bar{u}_2 \sigma^{\mu\nu} u_1 \int_0^1 dx dy dz \delta(x+y+z-1) z(z-1) \mathbb{X} \end{aligned} \quad (222)$$

$\mathbb{X}$ does not diverge.

$$\mathbb{X} \sim \int_1^\Lambda \frac{dq}{q^3} \sim -\frac{1}{2\Lambda^2} \sim 0, \quad \Lambda \rightarrow \infty \quad (223)$$

From Eq. B.39 in Schwartz

$$\int \frac{d^d p}{(p^2 - \Delta + i\epsilon)^3} = -\frac{i}{2} \pi^{d/2} \Delta^{d/2-3} \Gamma\left(\frac{6-d}{2}\right). \quad (224)$$

$$\mathbb{X} = -\frac{i\pi^2}{2\Delta} \Gamma(1), \quad \Gamma(1) = 1 \quad (225)$$

$$\mathbb{X} = -\frac{i\pi^2}{2\Delta} \quad (226)$$

$$\begin{aligned} \star &= 4e\odot + \frac{4e^3}{(2\pi)^4} \bar{u}_2 \gamma^\mu u_1 \int_0^1 dx dy dz \delta(x+y+z-1) \\ &\quad [m^2(1-4z+z^2) + (1-y)(1-x)p^2] \mathbb{X} \\ &\quad + \frac{me^3}{8\pi^2} p_\nu \bar{u}_2 \sigma^{\mu\nu} u_1 \int_0^1 dx dy dz \frac{\delta(x+y+z-1) z(1-z)}{\Delta} \end{aligned} \quad (227)$$

The fine structure constant is

$$\alpha = \frac{e^2}{4\pi} \implies e = 2\sqrt{\pi\alpha}. \quad (228)$$

$$\begin{aligned} \star &= 4e\odot + \frac{4e^3}{(2\pi)^4} \bar{u}_2 \gamma^\mu u_1 \int_0^1 dx dy dz \delta(x+y+z-1) \\ &\quad [m^2(1-4z+z^2) + (1-y)(1-x)p^2] \mathbb{X} \\ &\quad + m\sqrt{\frac{\alpha^3}{\pi}} p_\nu \bar{u}_2 \sigma^{\mu\nu} u_1 \int_0^1 dx dy dz \frac{\delta(x+y+z-1) z(1-z)}{\Delta} \end{aligned} \quad (229)$$

$$\Delta = (1-z)^2 m^2 - xyp^2 + z\lambda^2 \quad (230)$$

$$\begin{aligned} \star &= 4e\odot + \frac{4e^3}{(2\pi)^4} \bar{u}_2 \gamma^\mu u_1 \int_0^1 dx dy dz \delta(x+y+z-1) \\ &\quad [m^2(1-4z+z^2) + (1-y)(1-x)p^2] \mathbb{X} + p_\nu \bar{u}_2 \sigma^{\mu\nu} u_1 \mathbb{F} \end{aligned} \quad (231)$$

$$\mathbb{F} = m\sqrt{\frac{\alpha^3}{\pi}} \int_0^1 dx dy dz \frac{\delta(x+y+z-1) z(1-z)}{(1-z)^2 m^2 - xyp^2 + z\lambda^2} \quad (232)$$

Notice how the fictitious photon mass λ removes an IR divergence in $\mathbb{F}$. Consider the integrand of $\mathbb{F}$ with $\lambda = x = y = 0$ and $z = 1$ (we cannot have $x = y = z = 0$ because of the constraint $x + y + z = 1$):

$$\mathbb{F} \sim \frac{0}{0^2} = \frac{1}{0} = \infty. \quad (233)$$

That's why we need the photon mass!

$$\begin{aligned} \star &= 4e\odot - \frac{ie^3}{8\pi^2} \bar{u}_2 \gamma^\mu u_1 \int_0^1 dx dy dz \delta(x+y+z-1) \\ &\quad \frac{m^2(1-4z+z^2) + (1-y)(1-x)p^2}{\Delta} + p_\nu \bar{u}_2 \sigma^{\mu\nu} u_1 \mathbb{F} \end{aligned} \quad (234)$$

$$\begin{aligned} \star &= 4e\odot - i\sqrt{\frac{\alpha^3}{\pi}} \bar{u}_2 \gamma^\mu u_1 \int_0^1 dx dy dz \delta(x+y+z-1) \\ &\quad \frac{m^2(1-4z+z^2) + (1-y)(1-x)p^2}{(1-z)^2 m^2 - xyp^2 + z\lambda^2} + p_\nu \bar{u}_2 \sigma^{\mu\nu} u_1 \mathbb{F} \end{aligned} \quad (235)$$

$$\mathbb{F}_1 = \sqrt{\frac{\alpha^3}{\pi}} \int_0^1 dx dy dz \delta(x+y+z-1) \frac{m^2(1-4z+z^2) + (1-y)(1-x)p^2}{(1-z)^2 m^2 - xyp^2 + z\lambda^2} \quad (236)$$

$$\star = 4e\odot - i\bar{u}_2 \gamma^\mu u_1 \mathbb{F}_1 + p_\nu \bar{u}_2 \sigma^{\mu\nu} u_1 \mathbb{F} \quad (237)$$

$\mathbb{F}_1$ also has an IR divergence at $x = y = \lambda = 0$ & $z = 1$.

$$\mathbb{F}_1 \sim \frac{p^2 - 2m^2}{0^2} \sim \frac{1}{0} = \infty \quad (238)$$

$$\odot = \frac{e^2}{(2\pi)^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4 q \frac{\bar{u}_2 q^\alpha q^\beta \gamma_\alpha \gamma^\mu \gamma_\beta u_1}{(q^2 - \Delta + i\epsilon)^3} \quad (239)$$

$$\odot = \frac{e^2}{(2\pi)^4} \int_0^1 dx dy dz \delta(x+y+z-1) \bar{u}_2 \mathcal{B} \gamma_\alpha \gamma^\mu \gamma_\beta u_1 \quad (240)$$

$$\mathcal{B} = \int d^4 q \frac{q^\alpha q^\beta}{(q^2 - \Delta + i\epsilon)^3} \quad (241)$$

$\mathcal{B}$ is UV divergent.

$$\mathcal{B} \sim \int_1^\Lambda \frac{dq}{q} = \ln \Lambda, \quad \Lambda \rightarrow \infty \quad (242)$$

I regulate $\mathcal{B}$ with DimReg.

$$\odot(d) = \frac{e^2 \mu^{4-d}}{(2\pi)^d} \bar{u}_2 \gamma_\alpha \gamma^\mu \gamma_\beta u_1 \int_0^1 dx dy dz \delta(x+y+z-1) \mathcal{B}(d) \quad (243)$$

$$\mathcal{B}(d) = \int d^d q \frac{q^\alpha q^\beta}{(q^2 - \Delta + i\epsilon)^3} \quad (244)$$

$$\odot = \odot(4), \quad \mathcal{B} = \mathcal{B}(4) \quad (245)$$

I use Eq. 44.

$$\mathcal{B}(d) = \frac{\eta^{\alpha\beta}}{d} \int d^d q \frac{q^2}{(q^2 - \Delta + i\epsilon)^3} \quad (246)$$

I use Schwartz Eq. B.38

$$\int d^d p \frac{p^2}{(p^2 - \Delta + i\epsilon)^3} = \frac{i(2\pi)^d d}{4(4\pi)^{d/2} \Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right). \quad (247)$$

$$\mathcal{B}(d) = \frac{i\eta^{\alpha\beta}(2\pi)^d}{4(4\pi)^{d/2} \Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right) \quad (248)$$

$$\odot(d) = i \frac{e^2 \mu^{4-d}}{4(4\pi)^{d/2}} \bar{u}_2 \eta^{\alpha\beta} \gamma_\alpha \gamma^\mu \gamma_\beta u_1 \Gamma\left(\frac{4-d}{2}\right) \int_0^1 dx dy dz \frac{\delta(x+y+z-1)}{\Delta^{2-d/2}} \quad (249)$$

$$\odot(d) = i \frac{e^2 \mu^{4-d}}{4(4\pi)^{d/2}} \bar{u}_2 \gamma^\beta \gamma^\mu \gamma_\beta u_1 \Gamma\left(\frac{4-d}{2}\right) \int_0^1 dx dy dz \frac{\delta(x+y+z-1)}{\Delta^{2-d/2}} \quad (250)$$

$$\{\gamma^\beta, \gamma^\mu\} = 2\eta^{\beta\mu} \implies \gamma^\beta \gamma^\mu = 2\eta^{\beta\mu} - \gamma^\mu \gamma^\beta \quad (251)$$

$$\odot(d) = i \frac{e^2 \mu^{4-d}}{4(4\pi)^{d/2}} \bar{u}_2 (2\eta^{\beta\mu} \gamma_\beta - \gamma^\mu \gamma^\beta \gamma_\beta) u_1 \Gamma\left(\frac{4-d}{2}\right) \int_0^1 dx dy dz \frac{\delta(x+y+z-1)}{\Delta^{2-d/2}} \quad (252)$$

In d dimensions $\gamma^\alpha \gamma_\alpha = d$.

$$\odot(d) = i \frac{e^2 \mu^{4-d}}{4(4\pi)^{d/2}} \bar{u}_2 (2\gamma^\mu - \gamma^\mu d) u_1 \Gamma\left(\frac{4-d}{2}\right) \int_0^1 dx dy dz \frac{\delta(x+y+z-1)}{\Delta^{2-d/2}} \quad (253)$$

$$\odot(d) = i \frac{e^2 \mu^{4-d}}{4(4\pi)^{d/2}} (2-d) \bar{u}_2 \gamma^\mu u_1 \Gamma\left(\frac{4-d}{2}\right) \int_0^1 dx dy dz \frac{\delta(x+y+z-1)}{\Delta^{2-d/2}} \quad (254)$$

I return to 4D.

$$\odot = \odot(4) = \lim_{\epsilon \rightarrow 0} \odot(4-\epsilon) = \odot(4-\epsilon) \quad (255)$$

In the last step I make the limit $\epsilon \rightarrow 0$ implicit.

$$\odot = \frac{ie^2 \mu^\epsilon}{4(4\pi)^{2-\epsilon/2}} (\epsilon-2) \bar{u}_2 \gamma^\mu u_1 \Gamma\left(\frac{\epsilon}{2}\right) \int_0^1 dx dy dz \frac{\delta(x+y+z-1)}{\Delta^{\epsilon/2}} \quad (256)$$

$$\odot = \frac{ie^2 \mu^\epsilon (4\pi)^{\epsilon/2}}{4(4\pi)^2} (\epsilon-2) \bar{u}_2 \gamma^\mu u_1 \Gamma\left(\frac{\epsilon}{2}\right) \int_0^1 dx dy dz \frac{\delta(x+y+z-1)}{\Delta^{\epsilon/2}} \quad (257)$$

$$\odot = \frac{ie^2}{4(4\pi)^2} \bar{u}_2 \gamma^\mu u_1 \int_0^1 dx dy dz \delta(x+y+z-1) \left(2\sqrt{\frac{\pi}{\Delta}} \mu\right)^\epsilon (\epsilon-2) \Gamma\left(\frac{\epsilon}{2}\right) \quad (258)$$

I expand around $\epsilon = 0$.

$$\tilde{\mu}^2 = 4\pi e^{-\gamma} \mu^2 \quad (259)$$

$$\odot = \frac{ie^2}{4(4\pi)^2} \bar{u}_2 \gamma^\mu u_1 \int_0^1 dx dy dz \delta(x+y+z-1) \left[-\frac{4}{\epsilon} + 2 - 2 \ln\left(\frac{\tilde{\mu}^2}{\Delta}\right)\right] \quad (260)$$

$$\odot = -\frac{ie^2}{2(4\pi)^2} \bar{u}_2 \gamma^\mu u_1 \int_0^1 dx dy dz \delta(x+y+z-1) \left[\frac{2}{\epsilon} - 1 + \ln\left(\frac{\tilde{\mu}^2}{\Delta}\right)\right] \quad (261)$$

$$\star = 4e\odot - i\bar{u}_2 \gamma^\mu u_1 \mathbb{F}_1 + p_\nu \bar{u}_2 \sigma^{\mu\nu} u_1 \mathbb{F} \quad (262)$$

$$\star = -i\bar{u}_2 \gamma^\mu u_1 \mathbb{F}_2 - i\bar{u}_2 \gamma^\mu u_1 \mathbb{F}_1 + p_\nu \bar{u}_2 \sigma^{\mu\nu} u_1 \mathbb{F} \quad (263)$$

$$\mathbb{F}_2 = \frac{e^3}{8\pi^2} \int_0^1 dx dy dz \delta(x+y+z-1) \left[\frac{2}{\epsilon} - 1 + \ln\left(\frac{\tilde{\mu}^2}{\Delta}\right)\right] \quad (264)$$

$$\star = p_\nu \bar{u}_2 \sigma^{\mu\nu} u_1 \mathbb{F} - i(\mathbb{F}_2 + \mathbb{F}_1) \bar{u}_2 \gamma^\mu u_1 \quad (265)$$

$$\mathbb{F}_2 = \sqrt{\frac{\alpha^3}{\pi}} \int_0^1 dx dy dz \delta(x+y+z-1) \left[\frac{2}{\epsilon} - 1 + \ln \left(\frac{\tilde{\mu}^2}{\Delta} \right) \right] \quad (266)$$

$$\mathbb{F}_2 = \sqrt{\frac{\alpha^3}{\pi}} \int_0^1 dx dy dz \delta(x+y+z-1) \left[\frac{2}{\epsilon} - 1 + \ln \left(\frac{\tilde{\mu}^2}{(1-z)^2 m^2 - xyp^2 + z\lambda^2} \right) \right] \quad (267)$$

$\mathbb{F}_2$ also has an IR divergence at $x = y = \lambda = 0$ & $z = 1$.

$$\mathbb{F}_2 \sim 2 \ln \left(\frac{\tilde{\mu}}{0} \right) \sim \ln \infty \sim \infty \quad (268)$$

$$\star = p_\nu \bar{u}_2 \sigma^{\mu\nu} u_1 \mathbb{F} - i(\mathbb{F}_2 + \mathbb{F}_1) \bar{u}_2 \gamma^\mu u_1 \quad (269)$$

$$\mathbb{F} = \sqrt{\frac{\alpha^3}{\pi}} \mathcal{F} \quad (270)$$

$$\Delta = (1-z)^2 m^2 - xyp^2 + z\lambda^2 \quad (271)$$

$$\mathcal{F} = m \int_0^1 dx dy dz \delta(x+y+z-1) z \frac{1-z}{\Delta} \quad (272)$$

$$\mathbb{F}_1 = \sqrt{\frac{\alpha^3}{\pi}} \int_0^1 dx dy dz \delta(x+y+z-1) \frac{m^2(1-4z+z^2) + (1-y)(1-x)p^2}{\Delta} \quad (273)$$

$$\mathbb{F}_2 = \sqrt{\frac{\alpha^3}{\pi}} \int_0^1 dx dy dz \delta(x+y+z-1) \left[\frac{2}{\epsilon} - 1 + \ln \left(\frac{\tilde{\mu}^2}{\Delta} \right) \right] \quad (274)$$

$$\mathbb{F}_2 + \mathbb{F}_1 = \sqrt{\frac{\alpha^3}{\pi}} \mathcal{S} \quad (275)$$

$$\mathcal{S} = \frac{2}{\epsilon} - 1 + \int_0^1 dx dy dz \delta(x+y+z-1) \left[\frac{m^2(1-4z+z^2) + (1-y)(1-x)p^2}{\Delta} + \ln \left(\frac{\tilde{\mu}^2}{\Delta} \right) \right] \quad (276)$$

$$\star = \sqrt{\frac{\alpha^3}{\pi}} \bar{u}_2 (p_\nu \sigma^{\mu\nu} \mathcal{F} - i \mathcal{S} \gamma^\mu) u_1 \quad (277)$$

This disagrees with Eq. 19.54 in Schwartz. But given my careful derivation and checking I suspect Schwartz has a typo. That and the book has a lot of other typos.

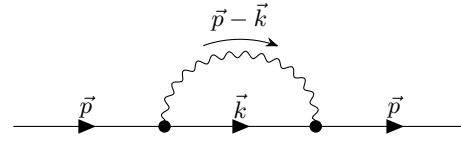


FIG. 5. Electron self energy Feynman diagram $\mathfrak{F}$. The external electrons are off-shell.

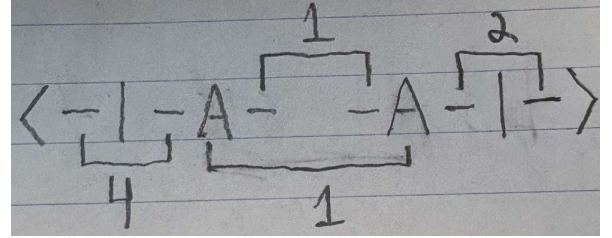


FIG. 6. Electron self energy symmetry factor combinatorics.

3. ELECTRON SELF ENERGY

I follow Schwartz Sec. 18.2. I show the Feynman diagram $\mathfrak{F}$ in Fig. 5 and the symmetry factor, S , combinatorics in Fig. 6.

$$\mathfrak{F} = \frac{S}{(2\pi)^4} \int d^4 k (-ie\gamma^\mu) i \frac{\not{k} + m}{k^2 - m^2 + i\epsilon} (-ie\gamma^\nu) \frac{-i\eta_{\mu\nu}}{(\vec{p} - \vec{k})^2 + i\epsilon} \quad (278)$$

$$S = \frac{4 \times 2}{(2!)^2 2!} = 1 \quad (279)$$

$$\mathfrak{F} = -\frac{e^2}{(2\pi)^4} \int d^4 k \frac{\gamma^\mu (\not{k} + m) \gamma^\nu \eta_{\mu\nu}}{(k^2 - m^2 + i\epsilon)[(\vec{p} - \vec{k})^2 + i\epsilon]} \quad (280)$$

$$\mathfrak{G} = -\frac{e^2}{(2\pi)^4} \int d^4k \frac{\gamma^\mu (\not{k} + m) \gamma_\mu}{(k^2 - m^2 + i\epsilon)[(\vec{p} - \vec{k})^2 + i\epsilon]} \quad (281)$$

I define

$$\mathfrak{G} = i\Sigma_2(\not{p}). \quad (282)$$

$$\Sigma_2(\not{p}) = \frac{ie^2}{(2\pi)^4} \int d^4k \frac{\gamma^\mu (\not{k} + m) \gamma_\mu}{(k^2 - m^2 + i\epsilon)[(\vec{p} - \vec{k})^2 + i\epsilon]} \quad (283)$$

I simplify the diagram with a Feynman parameter.

$$A = k^2 - m^2 + i\epsilon, \quad B = (\vec{p} - \vec{k})^2 + i\epsilon \quad (284)$$

$$\Sigma_2(\not{p}) = \frac{ie^2}{(2\pi)^4} \int d^4k \frac{\gamma^\mu (\not{k} + m) \gamma_\mu}{AB} \quad (285)$$

$$\frac{1}{AB} = \int_0^1 \frac{dx}{D^2}, \quad D = A + (B - A)x \quad (286)$$

$$\Sigma_2(\not{p}) = \frac{ie^2}{(2\pi)^4} \int d^4k \int_0^1 dx \frac{\gamma^\mu (\not{k} + m) \gamma_\mu}{D^2} \quad (287)$$

$$\Sigma_2(\not{p}) = \frac{ie^2}{(2\pi)^4} \int_0^1 dx \int d^4k \frac{\gamma^\mu (\not{k} + m) \gamma_\mu}{D^2} \quad (288)$$

$$D = k^2 - m^2 + i\epsilon + [(\vec{p} - \vec{k})^2 - k^2 + m^2]x \quad (289)$$

$$D = (k^2 - m^2)(1 - x) + (\vec{p} - \vec{k})^2 x + i\epsilon \quad (290)$$

$$\Sigma_2(\not{p}) = \frac{ie^2}{(2\pi)^4} \int_0^1 dx \int d^4k \frac{N}{D^2} \quad (291)$$

$$N = \gamma^\mu (\not{k} + m) \gamma_\mu = (\gamma^\mu \not{k} + m\gamma^\mu) \gamma_\mu \quad (292)$$

From Eq. 101

$$\gamma^\nu \not{p} = 2p^\nu - \not{p} \gamma^\nu. \quad (293)$$

$$\begin{aligned} N &= (2k^\mu - \not{k} \gamma^\mu + m\gamma^\mu) \gamma_\mu = 2k^\mu \gamma_\mu - \not{k} \gamma^\mu \gamma_\mu + m\gamma^\mu \gamma_\mu \\ &= 2\not{k} - \not{k} \gamma^\mu \gamma_\mu + m\gamma^\mu \gamma_\mu \end{aligned} \quad (294)$$

$$\gamma^\nu \gamma_\nu = 4I = 4 \quad (295)$$

I suppress the identity matrix in the last step.

$$N = 2\not{k} - 4\not{k} + 4m = 4m - 2\not{k} \quad (296)$$

$$\Sigma_2(\not{p}) = \frac{2ie^2}{(2\pi)^4} \int_0^1 dx \int d^4k \frac{2m - \not{k}}{D^2} \quad (297)$$

I complete the square.

$$D = k^2 - 2x\vec{p} \cdot \vec{k} + xp^2 - m^2(1 - x) + i\epsilon \quad (298)$$

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a} \quad (299)$$

$$a \rightarrow 1, \quad b \rightarrow -2x\vec{p}, \quad c \rightarrow xp^2 - m^2(1 - x) + i\epsilon, \quad x \rightarrow \vec{k} \quad (300)$$

$$D = (\vec{k} - x\vec{p})^2 - (m^2 - xp^2)(1 - x) + i\epsilon \quad (301)$$

$$\Delta = (m^2 - xp^2)(1 - x) \quad (302)$$

$$D = (\vec{k} - x\vec{p})^2 - \Delta + i\epsilon \quad (303)$$

$$\Sigma_2(\not{p}) = \frac{2ie^2}{(2\pi)^4} \int_0^1 dx \int d^4k \frac{2m - \not{k}}{[(\vec{k} - x\vec{p})^2 - \Delta + i\epsilon]^2} \quad (304)$$

I shift the integrand.

$$\vec{q} = \vec{k} - x\vec{p} \quad \vec{k} = \vec{q} + x\vec{p}, \quad d^4k = d^4q \quad (305)$$

$$\Sigma_2(\not{p}) = \frac{2ie^2}{(2\pi)^4} \int_0^1 dx \int d^4q \frac{2m - \not{q} - x\not{p}}{(q^2 - \Delta + i\epsilon)^2} \quad (306)$$

The term with one factor of q in the numerator is odd and integrates to zero.

$$\Sigma_2(\not{p}) = \frac{2ie^2}{(2\pi)^4} \int_0^1 dx \int d^4q \frac{2m - x\not{p}}{(q^2 - \Delta + i\epsilon)^2} \quad (307)$$

$$\Sigma_2(\not{p}) = \frac{2ie^2}{(2\pi)^4} \int_0^1 dx (2m - x\not{p}) \int \frac{d^4q}{(q^2 - \Delta + i\epsilon)^2} \quad (308)$$

$$\Sigma_2(\not{p}) = \frac{2ie^2}{(2\pi)^4} \int_0^1 dx (2m - x\not{p}) \mathbb{I} \quad (309)$$

$$\mathbb{I} = \int \frac{d^4q}{(q^2 - \Delta + i\epsilon)^2} \quad (310)$$

I use power counting to see if this integral diverges.

$$\mathbb{I} \sim \int_1^\Lambda \frac{dq}{q} = \ln \Lambda, \quad \Lambda \rightarrow \infty \quad (311)$$

It diverges logarithmically. I use dimensional regularization.

$$\Sigma_2(d, \not{p}) = \frac{2i\mu^{4-d}e^2}{(2\pi)^d} \int_0^1 dx (2m - x\not{p}) \mathbb{I}(d) \quad (312)$$

$$\mathbb{I}(d) = \int \frac{d^d q}{(q^2 - \Delta + i\epsilon)^2} \quad (313)$$

$$\Sigma_2(\not{p}) = \Sigma_2(4, \not{p}), \quad \mathbb{I} = \mathbb{I}(4) \quad (314)$$

In my Real Scalar Field Notes I derived

$$\int \frac{d^d p}{(p^2 - \Delta + i\epsilon)^2} = \frac{i\pi^{d/2}}{\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right). \quad (315)$$

$$\mathbb{I}(d) = \frac{i\pi^{d/2}}{\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right). \quad (316)$$

$$\Sigma_2(d, \not{p}) = \frac{2\pi^{d/2}e^2}{(2\pi)^d} \mu^{4-d} \Gamma\left(\frac{4-d}{2}\right) \int_0^1 dx \frac{x\not{p} - 2m}{\Delta^{2-d/2}} \quad (317)$$

$$\Sigma_2(\not{p}) = \lim_{\epsilon \rightarrow 0} \Sigma_2(4 - \epsilon, \not{p}) = \Sigma_2(4 - \epsilon, \not{p}) \quad (318)$$

Where the limit $\epsilon \rightarrow 0$ is implicit in the last equality.

$$\Sigma_2(\not{p}) = \frac{2\pi^{2-\epsilon/2}e^2}{(2\pi)^{4-\epsilon}} \mu^\epsilon \Gamma\left(\frac{\epsilon}{2}\right) \int_0^1 dx \frac{x\not{p} - 2m}{\Delta^{\epsilon/2}} \quad (319)$$

$$\Sigma_2(\not{p}) = \frac{2\pi^2 e^2 (2\pi)^\epsilon \mu^\epsilon}{(2\pi)^{4\pi^{1/2}} \Gamma\left(\frac{\epsilon}{2}\right)} \int_0^1 dx \frac{x\not{p} - 2m}{\Delta^{\epsilon/2}} \quad (320)$$

$$\Sigma_2(\not{p}) = \frac{e^2}{8\pi^2} (2\sqrt{\pi}\mu)^\epsilon \Gamma\left(\frac{\epsilon}{2}\right) \int_0^1 dx \frac{x\not{p} - 2m}{\Delta^{\epsilon/2}} \quad (321)$$

The fine structure constant is

$$\alpha = \frac{e^2}{4\pi}. \quad (322)$$

$$\Sigma_2(\not{p}) = \frac{\alpha}{2\pi} (2\sqrt{\pi}\mu)^\epsilon \Gamma\left(\frac{\epsilon}{2}\right) \int_0^1 dx \frac{x\not{p} - 2m}{\Delta^{\epsilon/2}} \quad (323)$$

I expand around $\epsilon = 0$.

$$\tilde{\mu}^2 = 4\pi e^{-\gamma} \mu^2 \quad (324)$$

Where $\gamma \approx 0.577216$ is the Euler–Mascheroni constant.

$$\Sigma_2(\not{p}) = \frac{\alpha}{2\pi} \int_0^1 dx (x\not{p} - 2m) \left[\frac{2}{\epsilon} + \ln\left(\frac{\tilde{\mu}^2}{\Delta}\right) \right] \quad (325)$$

$$\Sigma_2(\not{p}) = \frac{\alpha}{2\pi} \int_0^1 dx (x\not{p} - 2m) \left[\frac{2}{\epsilon} + \ln\left(\frac{\tilde{\mu}^2}{(m^2 - xp^2)(1-x)}\right) \right] \quad (326)$$